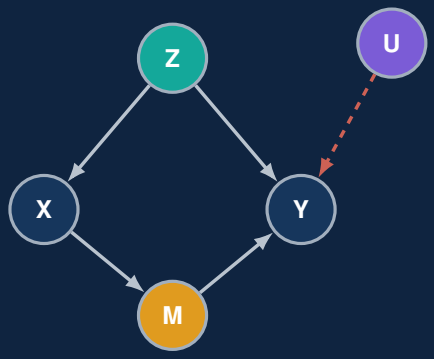


ADVANCED ARTIFICIAL INTELLIGENCE

Probabilistic Graphical Models and Causal Inference

Session 15

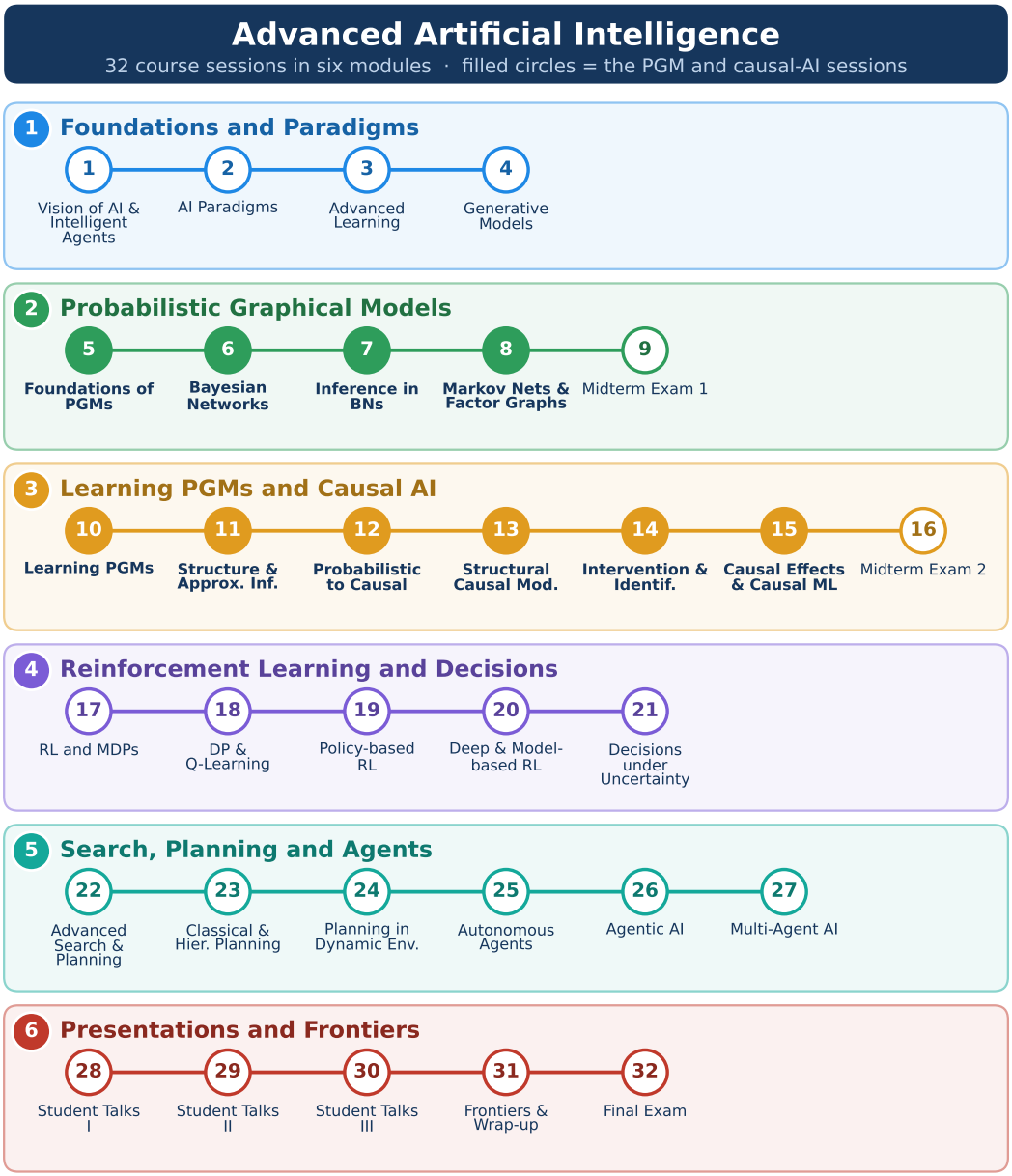
Causal Effects and Machine Learning

Lecture Notes

Instructor: Prof. M. Arashi

About these notes

These notes cover Session 15 of the course *Advanced Artificial Intelligence*, part of its probabilistic-graphical-model and causal-AI block (course sessions 5–15). The session follows a fixed structure: a roadmap and learning objectives, a "you are here" course map, the core theory organized in parts, a dedicated part on mathematical statistics, fully worked numerical and analytical examples, and a part connecting the topic to intelligent agents, agentic AI, reinforcement learning, multi-agent systems and the other sessions of the course. Every number is reproduced by the companion Python scripts (`python/s15_examples.py` for the numbers, `python/s15_figures.py` for the figures), and `questions/s15_questions.tex` contains ten questions with complete answers.



The Advanced Artificial Intelligence course. Filled circles mark the PGM and causal-AI sessions.

Contents

Session 15 Causal Effects and Machine Learning **1**

15.1 The Potential-Outcomes Framework . . . 3
15.2 Estimands and Identification . . . 4
15.3 Outcome Regression, Matching and Stratification . . . 6
15.4 Propensity Scores and Inverse Propensity Weighting . . . 7
15.5 Doubly Robust Estimation . . . 9
15.6 Heterogeneous Effects and Meta-Learners . . . 10
15.7 Causal Forests and Honesty . . . 12
15.8 Double/Debiased Machine Learning . . . 12
15.9 Uplift Modelling and Policy Learning . . . 13
15.10 Bias and Variance of Weighting Estimators . . . 14
15.11 Efficiency, Orthogonality and the Bias of AIPW . . . 16
15.12 Inference in Practice: Monte Carlo, Bootstrap and Overlap . . . 18
15.13 Worked Examples . . . 19
15.14 Off-Policy Evaluation in Reinforcement Learning . . . 23
15.15 Agentic AI: A/B Tests and Uplift Routing . . . 24
15.16 Multi-Agent AI: When SUTVA Fails . . . 25
15.17 Summary . . . 27
15.18 Appendix: Proofs . . . 28

SESSION 15 ADVANCED ARTIFICIAL INTELLIGENCE

Causal Effects and Machine Learning

ATE, treatment effects, counterfactuals, causal effect estimation and causal ML

Advanced Artificial Intelligence · Probabilistic Graphical Models and Causal Inference

Instructor: Prof. M. Arashi

PART I
Potential Outcomes & Estimands
SUTVA, ignorability, positivity; ATE, ATT, ATC, CATE, ITE; what is identified.

PART II
Estimating Average Effects
Regression, matching, stratification, propensity scores, IPW, AIPW.

PART III
Heterogeneity & Causal ML
Meta-learners, causal forests, double ML, uplift and policy learning.

PART IV
Mathematical Statistics
Unbiasedness, variance, efficiency bound, orthogonality, bootstrap, CLT.

PART V
AI Connections
Off-policy evaluation, agent A/B tests, interference among agents.

LEARNING OBJECTIVES

- State the potential-outcomes framework (SUTVA, consistency, ignorability, positivity) and derive identification formulas for the ATE, ATT, ATC and CATE; explain why individual effects are only *bounded*.
- Build and compare the classical estimators: outcome regression, matching, stratification, Horvitz–Thompson and Hájek IPW, and the doubly robust AIPW estimator.
- Estimate heterogeneous effects with S-, T-, X-, R- and DR-learners and causal forests, and turn them into targeting policies evaluated by Qini curves.
- Explain **double/debiased machine learning**: Neyman orthogonality and cross-fitting.
- Prove unbiasedness of IPW, derive its variance, state the semiparametric efficiency bound and the CLT for AIPW, and compute bootstrap confidence intervals.
- Recognise the same estimators in RL (importance-sampling and doubly robust off-policy evaluation), in A/B tests of agents, and in multi-agent systems where SUTVA fails.

Where we are. Sessions 5–11 (course sessions 5–8 and 10–11) built the probabilistic machinery; Session 12 (course 12) showed $P(y \mid x) \neq P(y \mid \mathrm{do}(x))$; Session 13 (course 13) introduced SCMs and counterfactuals; Session 14 (course 14) decided *when* a causal quantity is identified. This final session turns identification into *estimation* with finite data and machine learning. Next, the course moves to reinforcement learning (course sessions 17–20), where today's estimators reappear as off-policy evaluation, and later to agentic and multi-agent AI (course sessions 26–27).

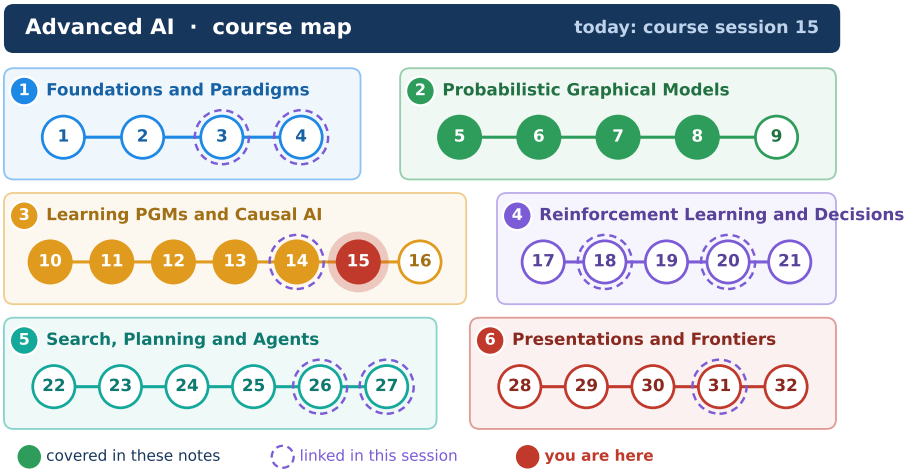


Figure 15.1. Where this session sits in the Advanced AI course. Dashed purple rings mark the course sessions we link to today.

OPENING MOTIVATION

Session 14 told us exactly which quantity to compute — $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z)P(z)$ under the backdoor criterion — but every worked example assumed the relevant conditional distributions were simply known. Real data is finite, noisy, and high-dimensional: Z might be dozens of covariates, not one categorical variable, and $P(y \mid x, z)$ is exactly the kind of function machine learning is built to estimate. This capstone session connects the identification theory of Sessions 12–14 to the practitioner's actual workflow: build estimators, using tools students already know (regression, classification, the reweighting ideas from Session 11's importance sampling), that target precisely the causal quantity Session 14 proved was well-defined.

Healthcare

Which patients benefit from a drug? Records are confounded by severity; we need the ATE *and* the CATE.

Online platforms

A coupon should go to customers it *changes*: uplift modelling targets $\widehat{\mathrm{CATE}}(z) > c$.

LLM agents

Is agent v2 (tools + reflection) worth its cost? Effects per task type drive a router.

PART I

Potential Outcomes and Estimands

What exactly do we want to estimate, and what can data reveal?

PART I · POTENTIAL OUTCOMES

15.1 The Potential-Outcomes Framework

Notation $T \in \{0,1\}$: treatment Z: covariates $Y(t)$: potential outcome $e(z) = P(T{=}1 \mid Z{=}z)$ $\mu_t(z) = \mathbb{E}[Y \mid T{=}t, Z{=}z]$ $\tau(z) = \mathrm{CATE}(z)$ n_1, n_0: numbers of treated and control units

Session 13 defined counterfactuals inside an SCM: $Y_x(u)$ is the value Y would take in the mutilated model M_x. Statistics and econometrics use the same object under the name *potential outcome* (Neyman–Rubin), with assumptions phrased directly on the potential outcomes instead of on a graph.

Definition 15.1 (Potential outcomes)**.** For a binary treatment, each unit i carries two potential outcomes $Y_i(1)$ (outcome if treated) and $Y_i(0)$ (outcome if untreated). The **individual treatment effect** is $\tau_i = Y_i(1) - Y_i(0)$. In SCM language $Y_i(t) = Y_{\mathrm{do}(T=t)}(u_i)$.

Definition 15.2 (Identifying assumptions)**.** A1. **SUTVA**: the potential outcomes of unit i depend only on i's own treatment (no interference) and there is a single version of each treatment level.

A2. **Consistency**: $Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0)$ — the observed outcome is the potential outcome of the treatment actually received.

A3. **Ignorability** (unconfoundedness): $(Y(0), Y(1)) \perp\!\!\!\perp T \mid Z$.

A4. **Positivity** (overlap): $0 < e(z) < 1$ for every z with positive density.

The fundamental problem: half of every row is missing

unit	Z_i	T_i	$Y_i(0)$	$Y_i(1)$	Y_i	τ_i
1	low	0	4.1	?	4.1	?
2	low	1	?	6.0	6.0	?
3	mid	0	5.2	?	5.2	?
4	mid	1	?	8.3	8.3	?
5	high	1	?	9.9	9.9	?
6	high	0	6.5	?	6.5	?

observed: $Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0)$

1. **SUTVA** — no interference; one version
2. **Consistency** — $T = t \Rightarrow Y = Y(t)$
3. **Ignorability** — $(Y(0), Y(1)) \perp T \mid Z$
4. **Positivity** — $0 < e(z) < 1$ for every z

Figure 15.2. The fundamental problem of causal inference: every unit reveals one potential outcome, so every individual effect τ_i is missing. The four assumptions on the right are what turns the missing-data problem into an estimable one.

Figure 15.2 shows why causal inference is a *missing-data problem*: the column τ_i is never observed. Ignorability says the missing half is "missing at random given Z"; positivity guarantees that both

halves are observed somewhere near every z.

> *Remark* 15.1 (Graphs and ignorability). If Z satisfies the back-door criterion in a causal DAG (Session 14), then $Y(t) \perp\!\!\!\perp T \mid Z$ (Pearl 2009, Ch. 11): the graph is how one *justifies* ignorability.

PART I · POTENTIAL OUTCOMES

15.2 Estimands and Identification

Suppose T is a binary treatment, Y the outcome, and Z a (possibly high-dimensional) set of covariates satisfying the backdoor criterion. The quantity of interest is usually the **average treatment effect**,

$$\mathrm{ATE} = \mathbb{E}\big[Y \mid \mathrm{do}(T{=}1)\big] - \mathbb{E}\big[Y \mid \mathrm{do}(T{=}0)\big] = \mathbb{E}_Z\Big[\mathbb{E}[Y \mid T{=}1, Z] - \mathbb{E}[Y \mid T{=}0, Z]\Big], \tag{15.1}$$

the second equality following directly from Session 14's backdoor adjustment theorem. But the ATE is only one of several estimands, each answering a different question (Table 15.1).

> **Definition 15.3** (Causal estimands)**.** With $\tau = Y(1) - Y(0)$: $\mathrm{ATE} = \mathbb{E}[\tau]$; $\mathrm{ATT} = \mathbb{E}[\tau \mid T{=}1]$ (effect on the treated); $\mathrm{ATC} = \mathbb{E}[\tau \mid T{=}0]$ (effect on the controls); $\mathrm{CATE}(z) = \mathbb{E}[\tau \mid Z{=}z]$; and the unit-level $\mathrm{ITE}_i = \tau_i$.

> **Theorem 15.1** (Identification under ignorability)**.** *Under SUTVA, consistency, ignorability and positivity,*
>
> $$\mathbb{E}[Y(t)] = \mathbb{E}\big[\mu_t(Z)\big] = \mathbb{E}\Big[\frac{\mathbb{1}\{T{=}t\}\,Y}{P(T{=}t \mid Z)}\Big], \qquad t \in \{0,1\}, \tag{15.2}$$
>
> $$\mathrm{CATE}(z) = \mu_1(z) - \mu_0(z), \qquad \mathrm{ATT} = \mathbb{E}[Y \mid T{=}1] - \mathbb{E}\big[\mu_0(Z) \mid T{=}1\big]. \tag{15.3}$$
>
> *The ATT needs only the weaker conditions $Y(0) \perp\!\!\!\perp T \mid Z$ and $e(z) < 1$.*

Proof. For the regression form, $\mathbb{E}[Y(t) \mid Z] = \mathbb{E}[Y(t) \mid T{=}t, Z] = \mathbb{E}[Y \mid T{=}t, Z] = \mu_t(Z)$ by ignorability and then consistency; conditioning on $\{T = t\}$ is legitimate because $P(T{=}t \mid Z) > 0$. Take expectations over Z. For the weighting form, $\mathbb{E}[\mathbb{1}\{T{=}t\}Y \mid Z] = \mathbb{E}[\mathbb{1}\{T{=}t\}Y(t) \mid Z] = P(T{=}t \mid Z)\,\mathbb{E}[Y(t) \mid Z]$, using consistency and then ignorability; divide by $P(T{=}t \mid Z)$ and average. The CATE follows by not averaging. For the ATT, $\mathbb{E}[Y(1) \mid T{=}1] = \mathbb{E}[Y \mid T{=}1]$ by consistency alone, and $\mathbb{E}[Y(0) \mid T{=}1] = \mathbb{E}\{\mathbb{E}[Y(0) \mid T{=}1, Z] \mid T{=}1\} = \mathbb{E}\{\mathbb{E}[Y(0) \mid T{=}0, Z] \mid T{=}1\} = \mathbb{E}[\mu_0(Z) \mid T{=}1]$, which uses only $Y(0) \perp\!\!\!\perp T \mid Z$ and $P(T{=}0 \mid Z) > 0$ on the support of $Z \mid T{=}1$. □

15.2.1 Unit-level counterfactuals: what even an RCT cannot reveal

The ITE τ_i depends on the *joint* distribution of $(Y(0), Y(1))$, while data — even a perfect randomized experiment — reveal only the two *marginals*. Hence the ITE, $\mathrm{Var}(\tau)$ and $P(\tau > 0)$ are not identified. For binary outcomes we can still bound them.

> **Proposition 15.1** (Fréchet bounds on benefit and harm)**.** *Let $p_1 = P(Y(1){=}1)$ and $p_0 = P(Y(0){=}1)$, so that $\mathrm{ATE} = p_1 - p_0$. The probability of benefit $P(Y(1){=}1, Y(0){=}0)$ is sharply*

bounded by $\max(0, p_1 - p_0) \le P(\textit{benefit}) \le \min(p_1, 1 - p_0)$*, and the probability of harm is* $P(Y(1){=}0, Y(0){=}1) = P(\textit{benefit}) - \text{ATE}$*. (These are the bounds on the probability of necessity and sufficiency of Session 13, read in potential-outcomes notation.)*

Proof. Write $P(a, b) = P(Y(1){=}a, Y(0){=}b)$ and let $b_{\text{ben}} = P(1, 0)$ be the benefit probability. The marginals force the other three cells of the 2×2 joint table: $P(1, 1) = p_1 - b_{\text{ben}}$ (from $P(1, 1) + P(1, 0) = p_1$), $P(0, 1) = p_0 - P(1, 1) = p_0 - p_1 + b_{\text{ben}}$ (harm), and $P(0, 0) = 1 - p_1 - P(0, 1) = 1 - p_0 - b_{\text{ben}}$. All four cells are ≥ 0 iff $b_{\text{ben}} \ge 0$, $b_{\text{ben}} \le p_1$, $b_{\text{ben}} \ge p_1 - p_0$ and $b_{\text{ben}} \le 1 - p_0$, which is the stated interval. Every value in that interval gives a valid joint distribution with the same marginals, so the bounds are sharp. Harm $= P(0, 1) = b_{\text{ben}} - (p_1 - p_0) = b_{\text{ben}} - \text{ATE}$. □

Worked Example 15.1 (Two worlds with the same RCT)**.** A hypothetical trial with a binary outcome ($Y = 1$: recovery) gives $p_1 = 0.6$, $p_0 = 0.4$, so ATE $= p_1 - p_0 = 0.6 - 0.4 = 0.20$. By Proposition 15.1, $P(\text{benefit}) \in [\max(0, 0.20), \min(0.6, 1 - 0.4)] = [0.20, 0.60]$ and $P(\text{harm}) = P(\text{benefit}) - 0.20 \in [0, 0.40]$. The two ends of the interval are two joint tables with the same marginals:

- *Monotone world* (benefit 0.20): harm $0.20 - 0.20 = 0$, always recover $P(1, 1) = 0.6 - 0.20 = 0.40$, never recover $P(0, 0) = 1 - 0.4 - 0.20 = 0.40$.
- *Polarised world* (benefit 0.60): harm $0.60 - 0.20 = 0.40$, always recover $0.6 - 0.60 = 0$, never recover $1 - 0.4 - 0.60 = 0$.

Check: in both worlds $P(Y(1){=}1) = P(1, 1) + P(1, 0) = 0.60$ and $P(Y(0){=}1) = P(1, 1) + P(0, 1) = 0.40$. Both worlds therefore produce identical trial data; deciding between them needs Session 13's structural assumptions (e.g. monotonicity, which forces the monotone world), not more data.

PYTHON LAB · `s15_examples.py` · `ex_probability_of_benefit()`

`P(benefit) in [0.20, 0.60], P(harm) in [0.00, 0.40]`; both tables give marginals $0.60, 0.40$.

Table 15.1. Estimands, their identification formulas and the assumptions each one needs.

Estimand	Identification formula	Needs
ATE	$\mathbb{E}[\mu_1(Z) - \mu_0(Z)] = \mathbb{E}\left[\frac{TY}{e(Z)} - \frac{(1-T)Y}{1-e(Z)}\right]$	full ignorability, $0 < e < 1$
ATT	$\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[\mu_0(Z) \mid T{=}1]$	$Y(0) \perp\!\!\!\perp T \mid Z$, $e < 1$
ATC	$\mathbb{E}[\mu_1(Z) \mid T{=}0] - \mathbb{E}[Y \mid T{=}0]$	$Y(1) \perp\!\!\!\perp T \mid Z$, $e > 0$
CATE(z)	$\mu_1(z) - \mu_0(z)$	ignorability and overlap at z
ITE	not identified; Fréchet bounds only	joint law of $(Y(0), Y(1))$

PART II

Estimating Average Treatment Effects

From identification formulas to estimators

PART II · AVERAGE EFFECTS

15.3 Outcome Regression, Matching and Stratification

Remark 15.2 (The running example: a confounded targeting study). Several results below are illustrated on one simulated study with known ground truth (`simulate_targeting()` in `s15_examples.py`, seed 2026). A firm has $n = 4000$ customers with engagement score $X \sim \text{Uniform}(0,1)$; X plays the role of the covariate Z. A targeted offer T is assigned with true propensity $e(x) = \text{sigmoid}(4x-2)$, where $\text{sigmoid}(u) = 1/(1+\exp(-u))$, so e ranges from $\text{sigmoid}(-2) = 0.119$ to $\text{sigmoid}(2) = 0.881$. The potential outcomes are

$$Y(0) = 1 + 2X + \varepsilon_0, \qquad Y(1) = Y(0) + (2 + 3X) + \varepsilon_1, \qquad \varepsilon_0, \varepsilon_1 \overset{\text{iid}}{\sim} N(0, 0.5^2),$$

hence $\mu_0(x) = 1 + 2x$, $\mu_1(x) = 3 + 5x$, $\tau(x) = \text{CATE}(x) = 2 + 3x$, $\sigma_0^2 = \text{Var}(\varepsilon_0) = 0.25$, $\sigma_1^2 = \text{Var}(\varepsilon_0 + \varepsilon_1) = 0.5$, and $\text{ATE} = \mathbb{E}[2 + 3X] = 2 + 3 \cdot 0.5 = 3.5$; in the drawn sample $\frac{1}{n}\sum_i (2 + 3X_i) = 3.506$ (the *sample* ATE). Engaged customers are both treated more often and have higher $Y(0)$, so X is a confounder. As a benchmark, the same customers also receive an independent coin-flip assignment $T^{\text{rct}} \sim \text{Bernoulli}(0.5)$ with outcome $Y(T^{\text{rct}})$. Nuisance models: $\hat{e}$ is a logistic regression of T on X; $\hat{\mu}_t$ come from one linear regression of Y on (X, T, XT) — both correctly specified. Monte Carlo studies redraw the data with $n = 1000$.

15.3.1 Outcome regression

Directly estimate $\hat{\mu}_1(z) = \widehat{\mathbb{E}}[Y \mid T{=}1, Z{=}z]$ and $\hat{\mu}_0(z) = \widehat{\mathbb{E}}[Y \mid T{=}0, Z{=}z]$ (any regression model — linear regression, gradient boosting, a neural network) using the observed data, and average their difference:

$$\widehat{\text{ATE}}_{\text{reg}} = \frac{1}{n}\sum_{i=1}^{n} \big[\hat{\mu}_1(z_i) - \hat{\mu}_0(z_i)\big]. \tag{15.4}$$

This is the most direct payoff of Session 10's learning machinery: it is ordinary supervised learning, applied twice (once per treatment arm), then combined via Session 14's adjustment formula. Its weakness: if $\hat{\mu}_1, \hat{\mu}_0$ are misspecified (wrong functional form), the ATE estimate is biased, with no diagnostic built in. In the running example (Remark 15.2) it gives $\widehat{\text{ATE}}_{\text{reg}} = 3.522$ (Worked Example 15.12).

COMMON PITFALL "Regress Y on T and Z and read off the coefficient of T" is *not* the ATE when effects are heterogeneous: it is a variance-weighted average of stratum effects (Worked Example 15.15).

15.3.2 Matching

Matching imputes each missing potential outcome by the observed outcome of a similar unit from the other arm. For the ATT, with $j(i) = \arg\min_{j:T_j=0} \|z_i - z_j\|$ the nearest control of treated unit i (controls may be reused: matching *with replacement*),

$$\widehat{\text{ATT}}_{\text{match}} = \frac{1}{n_1}\sum_{i:T_i=1} \big(Y_i - Y_{j(i)}\big).$$

Matching on the scalar $\hat{e}(z)$ (propensity-score matching) is justified by the balancing property below. With d continuous covariates the discrepancy $\|z_i - z_{j(i)}\|$ shrinks only like $n^{-1/d}$, so nearest-neighbour matching is biased in high dimension (Abadie and Imbens, 2006).

15.3.3 Stratification (subclassification)

Partition the propensity range into K strata (classically quintiles) and average within-stratum contrasts: $\hat{\tau}_{\text{strat}} = \sum_k (n_k/n)(\bar{Y}_{1k} - \bar{Y}_{0k})$, where n_k is the size of stratum k and $\bar{Y}_{tk}$ the mean outcome of its units with $T = t$. Five strata typically remove most confounding bias (about 90% for a single normally distributed covariate; Cochran, 1968); the rest comes from variation of $e(z)$ *within* strata.

Running example (Remark 15.2). Cutting $\hat{e}$ at its quintiles gives five strata of 800 customers each, so $n_k/n = 1/5$. The within-stratum differences are 2.390, 2.973, 3.533, 4.181, 4.789, whence $\hat{\tau}_{\text{strat}} = (2.390 + 2.973 + 3.533 + 4.181 + 4.789)/5 = 17.866/5 = 3.573$, against the sample ATE 3.506. The true mean CATEs of the five strata are 2.316, 2.917, 3.502, 4.093, 4.703: each stratum difference is slightly too large because $e(x)$ still increases with x inside each stratum — exactly the residual within-stratum confounding just described.

Worked Example 15.2 (Matching by hand)**.** Treated units (z, y): $(2, 9), (4, 12), (5, 13), (7.2, 16)$; controls: $(1, 5), (2, 6), (3, 7), (4.5, 9), (6, 10), (8, 12)$ (illustrative). Match each treated unit to the control with the smallest $|z_i - z_j|$, with replacement:

treated (z, y)	nearest control (z, y)	distance	runner-up distance	$y_i - y_{j(i)}$
$(2, 9)$	$(2, 6)$	0	1 (to $z = 1, 3$)	$9 - 6 = 3$
$(4, 12)$	$(4.5, 9)$	0.5	1 (to $z = 3$)	$12 - 9 = 3$
$(5, 13)$	$(4.5, 9)$	0.5	1 (to $z = 6$)	$13 - 9 = 4$
$(7.2, 16)$	$(8, 12)$	0.8	1.2 (to $z = 6$)	$16 - 12 = 4$

Hence $\widehat{\text{ATT}} = (3 + 3 + 4 + 4)/4 = 3.5$. The naive comparison uses the arm means $(9 + 12 + 13 + 16)/4 = 50/4 = 12.5$ and $(5 + 6 + 7 + 9 + 10 + 12)/6 = 49/6 = 8.167$, giving $12.5 - 8.167 = 4.333$; it is inflated because treated units have larger z, and z raises y.

PYTHON LAB · `s15_examples.py` · `ex_matching_toy()` and `ex_ps_stratification()`

```
matching ATT-hat = 3.500; naive = 12.500 - 8.167 = 4.333;  running example:
stratum sizes [800 800 800 800 800]; within-quintile differences [2.39 2.973
3.533 4.181 4.789]; stratified ATE = 3.573.
```

PART II · AVERAGE EFFECTS

15.4 Propensity Scores and Inverse Propensity Weighting

Definition 15.4 (Propensity score)**.** $e(z) = P(T{=}1 \mid Z{=}z)$, the probability of receiving treatment given covariates.

Theorem 15.2 (Balancing property)**.** *$T \perp\!\!\!\perp Z \mid e(Z)$, and consequently $Y(0), Y(1) \perp\!\!\!\perp T \mid e(Z)$ whenever $Y(0), Y(1) \perp\!\!\!\perp T \mid Z$ (the backdoor / "no unmeasured confounding" condition, in potential-outcomes notation). Adjusting for the single scalar $e(Z)$ suffices — adjusting for all of Z is sufficient but not necessary.*

Proof. Since $e(Z)$ is a function of Z, $P(T{=}1 \mid Z, e(Z)) = e(Z)$, while by the tower property $P(T{=}1 \mid e(Z)) = \mathbb{E}[\mathbb{E}[T \mid Z] \mid e(Z)] = e(Z)$. The two agree, so $T \perp\!\!\!\perp Z \mid e(Z)$. For the second claim, write $\mathbf{Y} = (Y(0), Y(1))$: $P(T{=}1 \mid \mathbf{Y}, e(Z)) = \mathbb{E}[P(T{=}1 \mid \mathbf{Y}, Z) \mid \mathbf{Y}, e(Z)] = \mathbb{E}[e(Z) \mid \mathbf{Y}, e(Z)] = e(Z)$, which does not depend on $\mathbf{Y}$ (Rosenbaum and Rubin, 1983). □

The propensity score also suggests a second family of estimators, built on the idea of Session 11's importance sampling: we cannot sample from the "everyone treated" world directly, so we reweight the treated units we *did* observe to stand in for that counterfactual population.

Definition 15.5 (Inverse propensity weighting)**.** The Horvitz–Thompson (HT) and the normalised Hájek estimators are

$$\widehat{\mathrm{ATE}}_{\mathrm{IPW}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{T_i Y_i}{\hat{e}(z_i)} - \frac{(1-T_i)Y_i}{1-\hat{e}(z_i)}\right], \qquad \widehat{\mathrm{ATE}}_{\mathrm{Háj}} = \frac{\sum_i T_i Y_i/\hat{e}_i}{\sum_i T_i/\hat{e}_i} - \frac{\sum_i (1-T_i)Y_i/(1-\hat{e}_i)}{\sum_i (1-T_i)/(1-\hat{e}_i)},$$

with the shorthand $\hat{e}_i = \hat{e}(z_i)$. When the true e is known it can be used in place of $\hat{e}$.

Each treated unit is reweighted by $1/\hat{e}(z_i)$ — units who were *unlikely* to be treated given their covariates but were treated anyway are up-weighted, since they are rare and informative "surprises" standing in for many similar untreated units; this is precisely Session 11's importance-sampling weight $w = p/q$, with the proposal q here being "what treatment assignment actually looked like" and the target p being "what it would look like if everyone were treated" (made exact in the box after Proposition 15.2). The weights of HT do not sum to n in a finite sample; Hájek divides by the realised sum, exactly like self-normalised importance sampling. Figure 15.3(b) shows the effect: after weighting, the covariate distribution of each arm matches the population.

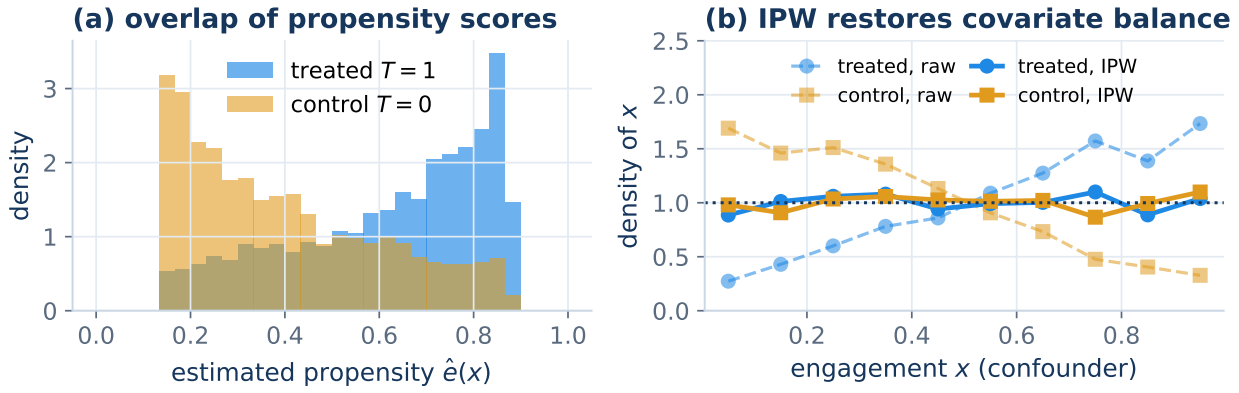


Figure 15.3. (a) Overlap of the estimated propensity scores $\hat{e}(x)$ of treated and control customers in the running example (Remark 15.2, $n = 4000$; $\hat{e} \in [0.135, 0.878]$). (b) Raw covariate densities differ by arm (confounding); IPW-weighted densities both match the population $U(0,1)$.

Worked Example 15.3 (A hospital study: six estimates from one table)**.** A hypothetical observational study has three severity strata. Sicker patients are treated more often and recover less:

stratum	n_z	true $e(z)$	treated n_{1z}	controls n_{0z}	$\bar{Y}_{1z}$	$\bar{Y}_{0z}$	$\hat{\tau}_z = \bar{Y}_{1z} - \bar{Y}_{0z}$
mild	100	0.2	24	76	9	8	1
moderate	60	0.5	27	33	7	5	2
severe	40	0.8	33	7	5	2	3

In total $n = 200$, $n_1 = 24 + 27 + 33 = 84$ and $n_0 = 76 + 33 + 7 = 116$.
1 *Naive:* treated mean $(24 \cdot 9 + 27 \cdot 7 + 33 \cdot 5)/84 = (216 + 189 + 165)/84 = 570/84 = 6.7857$, control mean $(76 \cdot 8 + 33 \cdot 5 + 7 \cdot 2)/116 = (608 + 165 + 14)/116 = 787/116 = 6.7845$, difference $6.7857 - 6.7845 = 0.0012$ — confounding hides the effect almost completely.
2 *Stratification* (= saturated outcome regression): $(100 \cdot 1 + 60 \cdot 2 + 40 \cdot 3)/200 = 340/200 = 1.700$.
3 $ATT = (24 \cdot 1 + 27 \cdot 2 + 33 \cdot 3)/84 = 177/84 = 2.107$ and $ATC = (76 \cdot 1 + 33 \cdot 2 + 7 \cdot 3)/116 = 163/116 = 1.405$: treated patients are sicker and benefit more.
4 *HT with the true e:* $\hat{\mathbb{E}}[Y(1)] = \frac{1}{200}\left(\frac{24 \cdot 9}{0.2} + \frac{27 \cdot 7}{0.5} + \frac{33 \cdot 5}{0.8}\right) = \frac{1080+378+206.25}{200} = \frac{1664.25}{200} = 8.321$ and $\hat{\mathbb{E}}[Y(0)] = \frac{1}{200}\left(\frac{76 \cdot 8}{0.8} + \frac{33 \cdot 5}{0.5} + \frac{7 \cdot 2}{0.2}\right) = \frac{760+330+70}{200} = \frac{1160}{200} = 5.800$, so $\widehat{\text{ATE}} = 8.321 - 5.800 = 2.521$. The treated weights sum to $24/0.2 + 27/0.5 + 33/0.8 = 120 + 54 + 41.25 = 215.25 \neq 200$ (control weights: $76/0.8 + 33/0.5 + 7/0.2 = 95 + 66 + 35 = 196$) because by chance more mild patients were treated (24) than e predicts $(100 \cdot 0.2 = 20)$.
5 *Hájek:* $1664.25/215.25 - 1160/196 = 7.732 - 5.918 = 1.813$, much closer to 1.700.
6 With the *empirical* propensities $n_{1z}/n_z = 24/100, 27/60, 33/40 = 0.24, 0.45, 0.825$, HT and Hájek both equal 1.700 exactly (Worked Example 15.4).

PYTHON LAB · s15_examples.py · ex_hospital_strata()

```
naive 0.0012; strat 1.700; ATT 2.107; ATC 1.405; HT 2.521 (weights 215.25,
196.00); Hajek 1.813; empirical e:  1.700, 1.700
```

Worked Example 15.4 (Estimated propensities turn IPW into stratification — analytical)**.** If Z is discrete and $\hat{e}(z) = n_{1z}/n_z$, then for every stratum $\sum_{i:z_i=z} T_i Y_i = n_{1z}\bar{Y}_{1z}$, so $\sum_{i:z_i=z} T_i Y_i/\hat{e}(z) = n_{1z}\bar{Y}_{1z} \cdot n_z/n_{1z} = n_z\bar{Y}_{1z}$, and $\sum_{i:z_i=z} T_i/\hat{e}(z) = n_{1z} \cdot n_z/n_{1z} = n_z$. In the same way, with $1-\hat{e}(z) = n_{0z}/n_z$, the control sums are $n_z\bar{Y}_{0z}$ and n_z. Hence both weight totals equal $\sum_z n_z = n$, and HT $= \frac{1}{n}\sum_z n_z(\bar{Y}_{1z} - \bar{Y}_{0z}) =$ Hájek $= \hat{\tau}_{\text{strat}}$. Estimating e — even when the true e is known — removes the chance imbalance seen in step 4. This is the finite-sample face of a deep fact: IPW with an *estimated* propensity is more efficient than with the true one (Hirano, Imbens and Ridder, 2003; in Table 15.3 the sd drops from 0.2670 to 0.0736).

PART II · AVERAGE EFFECTS

15.5 Doubly Robust Estimation

Theorem 15.3 (Double robustness)**.** *Define the augmented IPW (AIPW) estimator*

$$\widehat{\text{ATE}}_{AIPW} = \frac{1}{n}\sum_i \underbrace{\left[\hat{\mu}_1(z_i) - \hat{\mu}_0(z_i) + \frac{T_i(Y_i - \hat{\mu}_1(z_i))}{\hat{e}(z_i)} - \frac{(1-T_i)(Y_i - \hat{\mu}_0(z_i))}{1-\hat{e}(z_i)}\right]}_{\hat{\varphi}_i}. \tag{15.5}$$

This estimator is consistent for the true ATE if either *the outcome models $\hat{\mu}_1, \hat{\mu}_0$ are correctly*

specified or *the propensity model $\hat{e}$ is correctly specified — it needs only one of the two to be right, not both.*

INTUITION

Think of AIPW as the outcome-regression estimate plus a bias-correction term. If the outcome model is exactly right, the correction terms have expectation zero and add only noise. If the outcome model is wrong but the propensity model is right, the correction term's reweighting exactly cancels the outcome model's bias. "Two chances to get it right" is the standard, and accurate, shorthand.

The proof (Appendix, and the sharper Theorem 15.4 of Part IV) shows that the bias of AIPW is a *product* of the two models' errors. The pseudo-outcomes $\hat{\varphi}_i$ are useful well beyond the ATE: averaging them over subgroups, regressing them on Z (DR-learner), or summing them over a policy's choices (policy learning) all inherit double robustness.

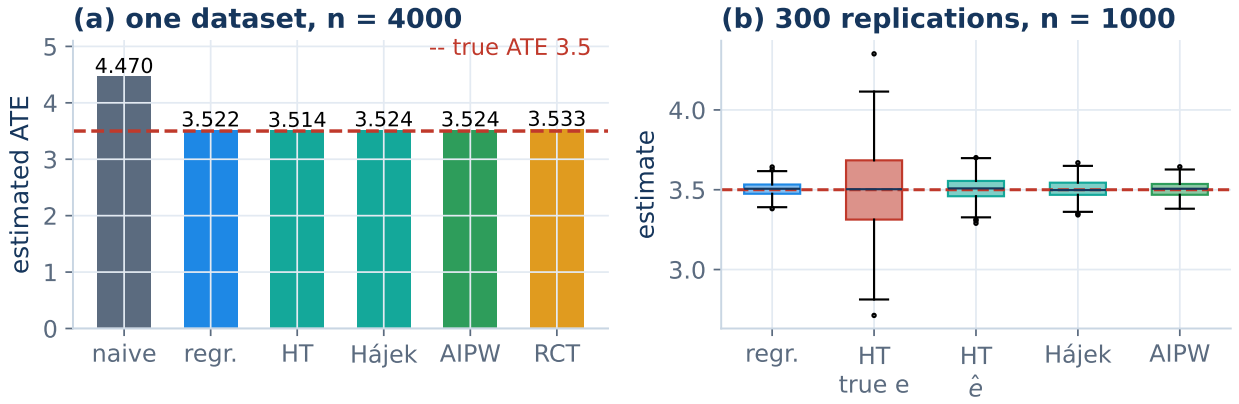


Figure 15.4. (a) On the $n = 4000$ running-example dataset (Remark 15.2 and Worked Example 15.12) all confounding-adjusted estimators recover the true ATE 3.5; the naive comparison does not. (b) Sampling distributions over 300 replications with $n = 1000$ (Table 15.3): IPW with the *true* propensity is unbiased but by far the noisiest; AIPW and regression are the most precise.

PART III

Heterogeneous Effects and Causal Machine Learning

From "does it work?" to "for whom, and whom should we treat?"

PART III · CAUSAL ML

15.6 Heterogeneous Effects and Meta-Learners

The ATE averages over the whole population, but a marketing team choosing whom to target, or a clinician choosing whom to treat, needs the **conditional average treatment effect**, $\mathrm{CATE}(z) = \mathbb{E}[Y \mid \mathrm{do}(T{=}1), Z{=}z] - \mathbb{E}[Y \mid \mathrm{do}(T{=}0), Z{=}z]$. *Meta-learners* repurpose ordinary supervised learning models to estimate $\mathrm{CATE}(z)$:

- **T-learner**: fit two separate models $\hat{\mu}_1(z)$, $\hat{\mu}_0(z)$ (one per treatment arm, exactly as in

outcome regression above) and set $\widehat{\mathrm{CATE}}(z) = \hat{\mu}_1(z) - \hat{\mu}_0(z)$.

- **S-learner**: fit a single model $\hat{\mu}(z, t)$ with treatment as an ordinary input feature, and set $\widehat{\mathrm{CATE}}(z) = \hat{\mu}(z, 1) - \hat{\mu}(z, 0)$.
- **X-learner**: impute individual-level "pseudo-effects" using the T-learner's cross-arm predictions, then regress these pseudo-effects on Z — generally more robust when treatment groups are very unbalanced in size. Concretely, $D_i^1 = Y_i - \hat{\mu}_0(z_i)$ for treated and $D_i^0 = \hat{\mu}_1(z_i) - Y_i$ for control units; regress D^1 on Z among the treated to get $\hat{\tau}_1$ and D^0 on Z among the controls to get $\hat{\tau}_0$, and combine $\hat{\tau}(z) = g(z)\hat{\tau}_0(z) + (1 - g(z))\hat{\tau}_1(z)$ with a weight $g \in [0, 1]$, usually $g = \hat{e}$: when few units are treated, $\hat{\tau}_1$ — built from $\hat{\mu}_0$, which is fitted on the large control arm — gets the larger weight $1 - \hat{e}$ (Künzel et al., 2019).
- **R-learner**: if $Y = \mu_0(Z) + T\tau(Z) + \varepsilon$ with $\mathbb{E}[\varepsilon \mid T, Z] = 0$, then $m(z) = \mathbb{E}[Y \mid Z{=}z] = \mu_0(z) + e(z)\tau(z)$, and subtracting gives Robinson's decomposition $Y - m(Z) = (T - e(Z))\tau(Z) + \varepsilon$. Hence $\hat{\tau} = \arg\min_\tau \sum_i \big[(Y_i - \hat{m}(z_i)) - (T_i - \hat{e}(z_i))\tau(z_i)\big]^2 + \Lambda(\tau)$, where Λ is a regularisation penalty on the function τ (Nie and Wager, 2021).
- **DR-learner**: regress the AIPW pseudo-outcomes $\hat{\varphi}_i$ of (15.5) on z_i. With the true nuisances, the pseudo-outcome $\varphi^\circ = \mu_1(Z) - \mu_0(Z) + T(Y - \mu_1(Z))/e(Z) - (1-T)(Y - \mu_0(Z))/(1 - e(Z))$ satisfies $\mathbb{E}[\varphi^\circ \mid Z] = \mu_1(Z) - \mu_0(Z) + 0 - 0 = \mathrm{CATE}(Z)$, because $\mathbb{E}[T(Y - \mu_1(Z)) \mid Z] = e(Z)\big(\mu_1(Z) - \mu_1(Z)\big) = 0$ (and likewise for the control term); so any regression method applies.

Table 15.2. Meta-learners for $\mathrm{CATE}(z)$.

Learner	Strength	Weakness
S	simple; pools data; good when effects are small	regularisation shrinks $\widehat{\mathrm{CATE}}$ towards 0 or a constant
T	flexible; no shared shrinkage	small arm ⇒ noisy; errors of two fits add
X	exploits the large arm; good for unbalanced designs	three stages; needs $\hat{e}$ for weighting
R	orthogonal loss; quasi-oracle rates	needs good $\hat{m}, \hat{e}$ and cross-fitting
DR	doubly robust pseudo-outcome	high variance where $\hat{e} \approx 0, 1$

Worked Example 15.5 (X-learner versus T-learner with 10% treated)**.** Simulate $n = 2000$ units (`xlearner_data()`, seed 11) with $Z \sim U(0,1)^5$, a complicated baseline $\mu_0(z) = 3\sin(\pi z_1 z_2) + 2(z_3 - 0.5)^2 + z_4$, a simple effect $\tau(z) = 1 + z_1$, treatment $T \sim \mathrm{Bernoulli}(0.1)$ independent of Z (the draw gives 210 treated, 1790 controls) and $Y = \mu_0(Z) + T\tau(Z) + \varepsilon$, $\varepsilon \sim N(0, 0.5^2)$. All base learners are random forests (150 trees, at least 5 units per leaf): the S-learner is one forest on (Z, T); the T-learner two forests, one per arm; the X-learner adds second-stage forests for D^1, D^0 and combines them with $g = e = 0.1$, i.e. $\hat{\tau} = 0.1\,\hat{\tau}_0 + 0.9\,\hat{\tau}_1$. Accuracy is measured on 1000 fresh test points (seed 12) by the root of the *precision in estimating heterogeneous effects*, $\sqrt{\mathrm{PEHE}} = \big(\frac{1}{1000}\sum_j (\hat{\tau}(z_j) - \tau(z_j))^2\big)^{1/2}$: it is 0.413 (S), 0.340 (T) and 0.152 (X). The T-learner must learn the complicated μ_0-shape twice, once from only 210 treated points; the X-learner borrows $\hat{\mu}_0$ from the 1790 controls and then only has to learn the simple τ (Fig. 15.5b).

PYTHON LAB · `s15_examples.py` · `ex_xlearner()`

```
S-learner:  root-PEHE = 0.413  T-learner:  0.340  X-learner:  0.152
```

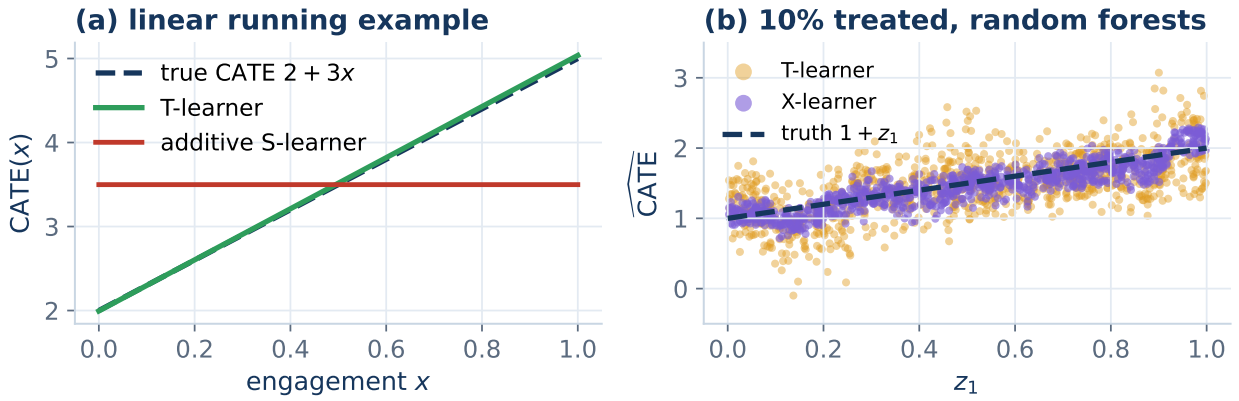


Figure 15.5. (a) Running example (Worked Example 15.13): the linear T-learner $\widehat{\mathrm{CATE}}(x) = 1.995 + 3.042x$ closely tracks the true $\mathrm{CATE}(x) = 2 + 3x$; an additive S-learner returns the constant 3.499. (b) Design of Worked Example 15.5, 1000 test points: with 10% treated, the X-learner (purple) is far less noisy than the T-learner (amber).

PART III · CAUSAL ML

15.7 Causal Forests and Honesty

A **causal tree** partitions Z-space into leaves with $\hat{\tau}(\text{leaf}) = \bar{Y}_{1,\text{leaf}} - \bar{Y}_{0,\text{leaf}}$, choosing splits that maximise the heterogeneity of $\hat{\tau}$ across leaves rather than the accuracy of predicting Y. **Honesty**: one subsample chooses the splits, a disjoint one fills the leaves, so a split cannot sit where noise created a large apparent effect. A **causal forest** averages honest trees built on subsamples; its estimates are asymptotically normal and centred at $\mathrm{CATE}(z)$ (Wager and Athey, 2018). Honesty is the tree-level cousin of cross-fitting: *never evaluate a fit on the data that chose it.*

PART III · CAUSAL ML

15.8 Double/Debiased Machine Learning

Regularisation bias of flexible ML predictors leaks into naive plug-in estimates; double/debiased ML (Chernozhukov et al., 2018) fixes this with orthogonal scores and cross-fitting. Consider the **partially linear model**

$$Y = \theta T + g(Z) + \varepsilon, \quad \mathbb{E}[\varepsilon \mid T, Z] = 0; \qquad T = m(Z) + v, \quad \mathbb{E}[v \mid Z] = 0. \tag{15.6}$$

Let $\ell(z) = \mathbb{E}[Y \mid Z{=}z]$. Taking $\mathbb{E}[\cdot \mid Z]$ of the outcome equation gives $\ell(Z) = \theta m(Z) + g(Z)$ (because $\mathbb{E}[T \mid Z] = m(Z)$ and $\mathbb{E}[\varepsilon \mid Z] = \mathbb{E}[\mathbb{E}(\varepsilon \mid T, Z) \mid Z] = 0$); subtracting, $Y - \ell(Z) = \theta\,(T - m(Z)) + \varepsilon$, so θ is the regression coefficient of the Y-residual on the T-residual (Frisch–Waugh–Lovell). The associated score $\psi = (Y - \ell(Z) - \theta(T - m(Z)))(T - m(Z))$ is **Neyman orthogonal**: small errors in $(\hat{\ell}, \hat{m})$ affect $\hat{\theta}$ only at second order (Definition 15.6).

Worked Example 15.6 (DML in a partially linear model)**.** Simulate (15.6) (`plm_data()`) with $\theta = 0.5$, $Z \sim N(0, I_5)$, nonlinear $m(z) = \cos z_1 + 0.5 z_2^2 + 0.5 \tanh z_3$ and $g(z) = 2 \sin z_1 + z_2^2 + 0.5 z_4$, independent noises $v, \varepsilon \sim N(0,1)$, $n = 1000$, ten replications (seeds 300–309). Nuisances $\hat{\ell}, \hat{m}$ are random forests of 40 trees with fully grown leaves (one unit per leaf); DML uses $K = 2$ folds. Means of the estimates over the ten replications: OLS of Y on T alone 1.035; OLS on T and linear Z 1.053 (the quadratic confounding through z_2^2 is missed); partialling-out

Algorithm 15.1 DML with K-fold cross-fitting (partially linear model)

Split $\{1,\dots,n\}$ into folds $I_1,\dots,I_K$.
for $k = 1,\dots,K$ **do**
    Fit $\hat{\ell}^{(-k)}$ and $\hat{m}^{(-k)}$ by any ML method on the data outside I_k.
    For $i \in I_k$: $\tilde{Y}_i = Y_i - \hat{\ell}^{(-k)}(Z_i)$, $\tilde{T}_i = T_i - \hat{m}^{(-k)}(Z_i)$.
end for
$\hat{\theta} = \sum_i \tilde{T}_i\tilde{Y}_i / \sum_i \tilde{T}_i^2$; $\hat{\psi}_i = (\tilde{Y}_i - \hat{\theta}\tilde{T}_i)\tilde{T}_i$; $\widehat{\mathrm{se}} = \sqrt{\frac{1}{n}\sum_i \hat{\psi}_i^2} / \left(\frac{1}{n}\sum_i \tilde{T}_i^2\right)/\sqrt{n}$.

with *in-sample* forest residuals 0.458 (overfitted residuals shrink the signal); **cross-fitted DML** 0.488 (sd 0.040 across replications). In the first replication DML gives $\hat{\theta} = 0.457$ and, from Algorithm 15.1, s.e. 0.040, so the 95% CI is $0.457 \pm 1.96 \times 0.040 = 0.457 \pm 0.078 = [0.379, 0.535]$, which covers 0.5.

PYTHON LAB · `s15_examples.py` · `ex_dml_plm()`

```
OLS 1.035; OLS+Z 1.053; in-sample 0.458; DML 0.488 (sd 0.040); rep. 1:
0.457, CI [0.379, 0.535]
```

PART III · CAUSAL ML

15.9 Uplift Modelling and Policy Learning

A treatment rule $\pi : z \mapsto \{0,1\}$ with per-unit cost c has net value $V(\pi) = \mathbb{E}[(\tau(Z) - c)\,\pi(Z)]$, maximised by the oracle rule $\pi^*(z) = \mathbb{1}\{\mathrm{CATE}(z) > c\}$. The plug-in rule "treat if $\widehat{\mathrm{CATE}}(z) > c$" is the bridge from causal ML to decisions; its value can be estimated from data by $\hat{V}(\pi) = \frac{1}{n}\sum_i \pi(z_i)(\hat{\varphi}_i - c)$ using the AIPW pseudo-outcomes, which inherits double robustness: its bias is a product of nuisance errors (Theorem 15.4).

Qini / uplift curves evaluate a CATE *ranking* on held-out randomized data: sort units by decreasing $\widehat{\mathrm{CATE}}$; after targeting the top fraction ϕ, the Qini value is $Q(\phi) = \sum_{\mathrm{top}} T_iY_i - \sum_{\mathrm{top}}(1 - T_i)Y_i \cdot n_T(\phi)/n_C(\phi)$, where the sums run over the targeted top fraction and $n_T(\phi), n_C(\phi)$ are the numbers of treated and control units in it: the control outcomes are rescaled to the size of the treated group, so $Q(\phi)$ is the incremental outcome generated. A useless model gives the straight line from 0 to $Q(1)$; the area between the curves is the Qini coefficient.

Worked Example 15.7 (Targeting with a cost)**.** In a hypothetical randomized campaign (`uplift_data()`, seed 21; $n = 6000$, the first 3000 for training, the last 3000 for testing), $Z \sim U(0,1)^2$, $T \sim \mathrm{Bernoulli}(0.5)$, $Y = 1 + Z_2 + T\tau(Z) + N(0,1)$ with $\tau(z) = 2z_1 - 0.8$ (the offer *hurts* customers with $z_1 < 0.4$), and cost $c = 0.3$ in outcome units. A T-learner with one linear regression per arm gives $\widehat{\mathrm{CATE}}(z) = -0.709 + 1.846z_1 + 0.007z_2$. On the test half the Qini curve peaks at 552.7 when 62% are targeted, against 259.3 for targeting everyone (Fig. 15.6a). Policy values are computed on the 3000 test customers with their *true* $\tau(z)$ (possible only in a simulation) as $\frac{1}{3000}\sum_j(\tau(z_j) - 0.3)\pi(z_j)$. The policy "treat if $\widehat{\mathrm{CATE}} > 0.3$" treats 45.8% and earns +0.2014 per user — the same, to four decimals, as the oracle rule "treat if $\tau > 0.3$" on these customers (which treats 45.1%) — against −0.0951 for treating everyone.

In the population the oracle treats exactly when $2z_1 - 0.8 > 0.3$, i.e. $z_1 > 0.55$, and earns

$$\mathbb{E}[(2Z_1 - 1.1)^+] = \int_{0.55}^{1} (2z - 1.1)\, dz = \big[z^2 - 1.1z\big]_{0.55}^{1} = (1 - 1.1) - (0.3025 - 0.605) = 0.2025.$$

PYTHON LAB · s15_examples.py · ex_uplift_policy()

```
Qini peak 552.7 at 62%; treat all -0.0951; CATE-hat > 0.3:  share 0.458,
+0.2014; oracle (test) 0.451, +0.2014; population oracle 0.2025
```

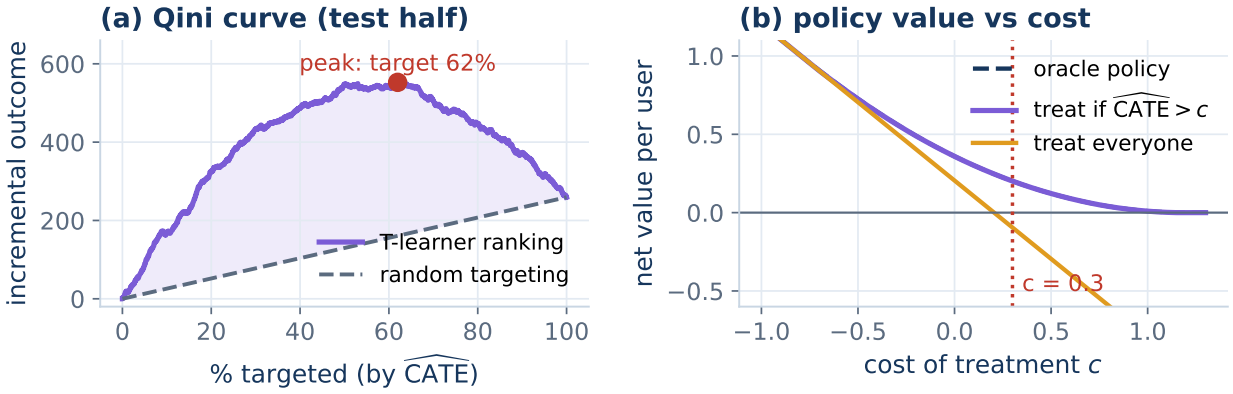


Figure 15.6. (a) Qini curve of the T-learner ranking against random targeting. (b) Net value of three policies as the treatment cost varies: acting on $\widehat{\mathrm{CATE}}$ is almost as good as the oracle.

PART IV

Mathematical Statistics of Effect Estimators

Bias, variance, efficiency, orthogonality and inference

PART IV · MATHEMATICAL STATISTICS

15.10 Bias and Variance of Weighting Estimators

Proposition 15.2 (IPW is unbiased; its variance)**.** *Let (Z_i, T_i, Y_i) be i.i.d., the assumptions of Definition 15.2 hold and e be* known. *Then the HT estimator is unbiased, $\mathbb{E}[\widehat{\mathrm{ATE}}_{IPW}] = \mathrm{ATE}$, and*

$$n\,\mathrm{Var}\big(\widehat{\mathrm{ATE}}_{IPW}\big) = \mathbb{E}\Big[\frac{Y(1)^2}{e(Z)} + \frac{Y(0)^2}{1 - e(Z)}\Big] - \mathrm{ATE}^2.$$

Proof. Write $\xi = TY/e(Z) - (1-T)Y/(1-e(Z))$ for one unit's term, so $\widehat{\mathrm{ATE}}_{\mathrm{IPW}} = \bar{\xi}$, the mean of n i.i.d. copies. By consistency $TY = TY(1)$, and by ignorability $\mathbb{E}[TY(1) \mid Z] = \mathbb{E}[T \mid Z]\,\mathbb{E}[Y(1) \mid Z] = e(Z)\mathbb{E}[Y(1) \mid Z]$, so $\mathbb{E}[TY/e(Z)] = \mathbb{E}\big[\mathbb{E}[Y(1) \mid Z]\big] = \mathbb{E}[Y(1)]$; similarly for the control term, hence $\mathbb{E}\xi = \mathrm{ATE}$ and $\bar{\xi}$ is unbiased. For the variance, $T(1-T) = 0$ kills the cross term in ξ^2, so $\xi^2 = TY(1)^2/e(Z)^2 + (1-T)Y(0)^2/(1-e(Z))^2$, and conditioning on Z as before gives $\mathbb{E}\xi^2 = \mathbb{E}[Y(1)^2/e(Z) + Y(0)^2/(1-e(Z))]$. Then $\mathrm{Var}(\bar{\xi}) = \mathrm{Var}(\xi)/n = (\mathbb{E}\xi^2 - \mathrm{ATE}^2)/n$. □

The variance contains $Y(t)^2$, not $\mathrm{Var}(Y(t) \mid Z)$, so HT is not location-invariant, and it blows up as $e(Z) \to 0$ or 1. In the running example (Remark 15.2), $\mathbb{E}[Y(1)^2 \mid X] = \sigma_1^2 + \mu_1(X)^2 = 0.5 + (3+5X)^2$ and $\mathbb{E}[Y(0)^2 \mid X] = 0.25 + (1+2X)^2$; numerical integration over $X \sim U(0,1)$ gives $\mathbb{E}[(0.5 + (3+5X)^2)/e(X)] = 67.045$ and $\mathbb{E}[(0.25 + (1+2X)^2)/(1-e(X))] = 17.027$, so $n\mathrm{Var} = 67.045 + 17.027 - 3.5^2 = 84.072 - 12.25 = 71.822$ per observation, i.e. sd $\sqrt{71.822/1000} = 0.2680$ at $n = 1000$, matching the Monte Carlo sd 0.2670 in Table 15.3.

MATHEMATICAL STATISTICS · IPW as importance sampling, and why Hájek helps

$\mathbb{E}[Y(1)] = \mathbb{E}_p[Y]$ under the "all treated" law p (draw Z, set $T = 1$), while the data come from the observational law q (draw Z, then $T \sim \mathrm{Bernoulli}(e(Z))$). The density ratio is $w = p/q = T/e(Z)$, so HT is the importance-sampling estimator $\hat{\mu}_{\mathrm{IS}}$ and Hájek the self-normalised one $\hat{\mu}_{\mathrm{SNIS}}$ of Session 11. By the delta method the Hájek estimator of $\mathbb{E}[Y(1)]$ has asymptotic variance $\mathbb{E}[(Y(1) - \mathbb{E}Y(1))^2/e(Z)]/n$ — the outcome is *centred*, so the variance no longer depends on the location of Y. Its small $O(1/n)$ ratio bias is the price.

Proposition 15.3 (Neyman variance in a randomized experiment)**.** *In an RCT with n_1 treated and n_0 controls drawn i.i.d. from a super-population, the difference in means $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$ is unbiased with $\mathrm{Var}(\hat{\tau}) = \sigma_1^2/n_1 + \sigma_0^2/n_0$, $\sigma_t^2 = \mathrm{Var}(Y(t))$, estimated by $s_1^2/n_1 + s_0^2/n_0$.*

Proof. Randomization gives $T \perp\!\!\!\perp (Y(0), Y(1))$, so the treated outcomes are an i.i.d. sample of $Y(1)$ and the controls of $Y(0)$, independent of each other given the arm sizes. Hence $\mathbb{E}\bar{Y}_t = \mathbb{E}Y(t)$, $\mathrm{Var}\bar{Y}_t = \sigma_t^2/n_t$ and the covariance is 0. (For a fixed finite population of n units with randomly assigned treatment, the estimator $s_1^2/n_1 + s_0^2/n_0$ is conservative: the true variance subtracts S_τ^2/n, where S_τ^2 is the finite-population variance of the unit effects τ_i, which is unidentifiable because it involves the joint law of $(Y(0), Y(1))$.) □

Worked Example 15.8 (A confidence interval for an RCT)**.** Hypothetical summary data: $n_1 = 60$, $\bar{y}_1 = 7.4$, $s_1 = 2.0$; $n_0 = 40$, $\bar{y}_0 = 6.1$, $s_0 = 1.5$. Then $\hat{\tau} = 7.4 - 6.1 = 1.30$, $\mathrm{SE} = \sqrt{s_1^2/n_1 + s_0^2/n_0} = \sqrt{4/60 + 2.25/40} = \sqrt{0.06667 + 0.05625} = \sqrt{0.12292} = 0.3506$, and the 95% CI is $1.30 \pm 1.96 \times 0.3506 = 1.30 \pm 0.687 = [0.613, 1.987]$.

The coin-flip arm of the running example (Remark 15.2) has $n_1 = 1987$, $n_0 = 2013$, $\bar{y}_1 = 5.522$, $\bar{y}_0 = 1.990$, $s_1^2 = 2.676$, $s_0^2 = 0.581$, so $\hat{\tau} = 3.533$ (difference of the unrounded means), $\mathrm{SE} = \sqrt{2.676/1987 + 0.581/2013} = \sqrt{0.001347 + 0.000288} = 0.0404$, and the CI is $3.533 \pm 1.96 \times 0.0404 = [3.453, 3.612]$.

PYTHON LAB · s15_examples.py · ex_neyman_rct()

```
tau-hat 1.30, SE 0.3506, CI [0.613, 1.987]; running-example RCT: n1 1987, n0
2013, s1^2 2.676, s0^2 0.581; 3.533 +/- 1.96*0.0404 -> [3.453, 3.612]
```

PART IV · MATHEMATICAL STATISTICS

15.11 Efficiency, Orthogonality and the Bias of AIPW

MATHEMATICAL STATISTICS · Efficient influence function and the semiparametric bound

In the nonparametric model for (Z, T, Y) the efficient influence function of $\psi = \text{ATE}$ is (Hahn, 1998)

$$\varphi(Z,T,Y) = \mu_1(Z) - \mu_0(Z) + \frac{T(Y-\mu_1(Z))}{e(Z)} - \frac{(1-T)(Y-\mu_0(Z))}{1-e(Z)} - \psi,$$

so no regular estimator has asymptotic variance below

$$V_{\text{eff}} = \text{Var}(\varphi) = \mathbb{E}\Big[\frac{\sigma_1^2(Z)}{e(Z)} + \frac{\sigma_0^2(Z)}{1-e(Z)} + \big(\tau(Z)-\psi\big)^2\Big], \qquad \sigma_t^2(z) = \text{Var}(Y \mid T{=}t, Z{=}z).$$

AIPW is the sample mean of $\varphi + \psi$ with estimated nuisances, which is why it attains V_{eff} (Theorem 15.5). The bound makes the role of overlap explicit: it is driven by $\mathbb{E}[1/e(Z)]$ and $\mathbb{E}[1/(1-e(Z))]$.

Worked Example 15.9 (The bound in the running example)**.** From Remark 15.2: $\sigma_1^2 = 0.5$, $\sigma_0^2 = 0.25$ (constant in x), $\tau(x) = 2 + 3x$ with $\text{Var}\tau(X) = 9\text{Var}X = 9/12 = 0.75$, and $e(x) = \text{sigmoid}(4x{-}2)$. Since $1/\,\text{sigmoid}(u) = 1{+}\exp(-u)$ and $1/(1{-}\text{sigmoid}(u)) = 1{+}\exp(u)$,

$$\mathbb{E}\Big[\frac{1}{e(X)}\Big] = \int_0^1 \big(1{+}\exp(2{-}4x)\big)dx = 1{+}\frac{\exp(2)-\exp(-2)}{4} = 1{+}1.8134 = 2.8134 = \mathbb{E}\Big[\frac{1}{1-e(X)}\Big],$$

so $\mathbb{E}[0.5/e{+}0.25/(1{-}e)] = 0.75{\times}2.8134 = 2.1101$ and $V_{\text{eff}} = 2.1101{+}0.75 = 2.8601$. No regular estimator can beat sd $\sqrt{2.8601/1000} = 0.0535$ at $n = 1000$ or $\sqrt{2.8601/4000} = 0.0267$ at $n = 4000$. HT with the true e has sd 0.2680 at $n = 1000$ (computed just below Proposition 15.2), $0.2680/0.0535 = 5.0$ times the bound.

The Monte Carlo sd of AIPW in Table 15.3 is 0.0502, slightly *below* the bound. This is Monte Carlo error, not a contradiction: an sd estimated from $R = 300$ replications has standard error about $\text{sd}/\sqrt{2(R-1)} = 0.0021$, and a longer run with $R = 2000$ replications (`ex_bound_check()`) gives 0.0534 against the bound 0.0535, with CI coverage 0.946. Over the 300 replications the mean influence-function s.e. of AIPW is 0.0535 — it estimates the bound — and on the $n = 4000$ dataset it is 0.0258 (Worked Example 15.10). AIPW sits at the bound. (Outcome regression with its correctly specified *linear* model has sd 0.0486, and 0.0521 with $R = 2000$: below the bound, which is possible because it is not a regular estimator in the nonparametric model — it is consistent only when μ_t really is linear.)

PYTHON LAB · s15_examples.py · ex_efficiency_bound()

```
E[1/e] = 2.8134; V = 2.1101 + 0.75 = 2.8601; bounds 0.0535 / 0.0267; HT
known e:  67.045 + 17.027 - 12.25 = 71.822, sd 0.2680
ex_bound_check():  kappa 1.0, R = 2000:  sd AIPW 0.0534, bound 0.0535, sd
regression 0.0521, coverage 0.946
```

Theorem 15.4 (The bias of AIPW is a product of errors)**.** *Let $\hat{\mu}_1, \hat{e}$ be* fixed *functions (e.g. fitted on an independent fold) with $\hat{e} > 0$. Then*

$$\mathbb{E}\Big[\hat{\mu}_1(Z) + \frac{T(Y - \hat{\mu}_1(Z))}{\hat{e}(Z)}\Big] - \mathbb{E}[Y(1)] = \mathbb{E}\Big[\big(\hat{\mu}_1(Z) - \mu_1(Z)\big)\,\frac{\hat{e}(Z) - e(Z)}{\hat{e}(Z)}\Big],$$

and analogously for the control arm. Consequently, if $\hat{e} \geq \varepsilon > 0$, $|bias| \leq \varepsilon^{-1} \|\hat{\mu}_1 - \mu_1\|_2 \, \|\hat{e} - e\|_2$.

Proof. Condition on Z: $\mathbb{E}[T(Y - \hat{\mu}_1(Z)) \mid Z] = e(Z)(\mu_1(Z) - \hat{\mu}_1(Z))$ by ignorability and consistency. So the left-hand mean equals $\mathbb{E}[\hat{\mu}_1 + e(\mu_1 - \hat{\mu}_1)/\hat{e}]$; subtracting $\mathbb{E}[Y(1)] = \mathbb{E}[\mu_1(Z)]$ leaves $\mathbb{E}[(\hat{\mu}_1 - \mu_1)(1 - e/\hat{e})] = \mathbb{E}[(\hat{\mu}_1 - \mu_1)(\hat{e} - e)/\hat{e}]$. The inequality is Cauchy–Schwarz. □

Corollary 15.1 (Double robustness and rate double robustness)**.** *The bias vanishes if $\hat{\mu}_1 = \mu_1$ or $\hat{e} = e$ (Theorem 15.3). More importantly, if both nuisances converge at rate $o_p(n^{-1/4})$ in L_2 — slow rates achievable by random forests, boosting or neural nets — the bias is $o_p(n^{-1/2})$ and does not affect $\sqrt{n}$-inference.*

Definition 15.6 (Neyman orthogonality)**.** Let $W = (Z, T, Y)$ denote one observation, θ the target parameter with true value θ_0, and η the nuisance functions (e.g. $\eta = (\mu_0, \mu_1, e)$) with true value η_0. A score $\psi(W; \theta, \eta)$ with $\mathbb{E}\psi(W; \theta_0, \eta_0) = 0$ is **Neyman orthogonal** if for every direction $\eta - \eta_0$, $\frac{\partial}{\partial r}\mathbb{E}\,\psi\big(W; \theta_0, \eta_0 + r(\eta - \eta_0)\big)\big|_{r=0} = 0$.

Proposition 15.4 (The AIPW score is orthogonal; the plug-in score is not)**.** *For $\psi_{AIPW} = \mu_1(Z) + T(Y - \mu_1(Z))/e(Z) - \theta$ (target $\theta = \mathbb{E}[Y(1)]$) the Gâteaux derivatives in both nuisance directions are zero, whereas the regression score $\psi_{reg} = \mu_1(Z) - \theta$ has derivative $\mathbb{E}[\delta(Z)] \neq 0$ in direction δ.*

Proof. All expectations are evaluated by first conditioning on Z, using $\mathbb{E}[T \mid Z] = e(Z)$ and $\mathbb{E}[T(Y - \mu_1(Z)) \mid Z] = e(Z)(\mu_1(Z) - \mu_1(Z)) = 0$ (ignorability and consistency). Perturb $\mu_1 \to \mu_1 + r\delta$: the score becomes $\psi_{\text{AIPW}} + r\delta(Z)(1 - T/e(Z))$, so the derivative is $\mathbb{E}[\delta(Z)(1 - T/e(Z))] = \mathbb{E}[\delta(Z)(1 - e(Z)/e(Z))] = 0$. Perturb $e \to e + r\zeta$: $\frac{d}{dr}\frac{1}{e + r\zeta}\big|_{r=0} = -\zeta/e^2$, so the derivative is $-\mathbb{E}[T(Y - \mu_1(Z))\zeta(Z)/e(Z)^2] = -\mathbb{E}[\zeta(Z)e(Z)(\mu_1(Z) - \mu_1(Z))/e(Z)^2] = 0$. For ψ_{reg} the derivative is $\mathbb{E}\delta(Z)$, which is non-zero for a generic error δ: first-order nuisance error becomes first-order bias. □

! Orthogonality makes nuisance errors enter only through products (Theorem 15.4); cross-fitting makes $\hat{\mu}, \hat{e}$ independent of the points they are evaluated on. Together they license any ML method for the nuisances.

Theorem 15.5 (CLT for cross-fitted AIPW)**.** *Assume i.i.d. data, the four identifying assumptions, strict overlap $\varepsilon \leq e \leq 1 - \varepsilon$, bounded conditional variances, cross-fitted nuisances that are consistent in L_2 with $\|\hat{\mu}_t - \mu_t\|_2 \, \|\hat{e} - e\|_2 = o_p(n^{-1/2})$, and $\varepsilon \leq \hat{e} \leq 1 - \varepsilon$. Then*

$$\sqrt{n}\,\big(\widehat{\text{ATE}}_{AIPW} - \text{ATE}\big) \xrightarrow{d} N(0, V_{\text{eff}}), \qquad \hat{V} = \tfrac{1}{n}\textstyle\sum_i (\hat{\varphi}_i - \widehat{\text{ATE}})^2 \xrightarrow{p} V_{\text{eff}}.$$

Proof sketch in the Appendix: $\widehat{\text{ATE}} - \text{ATE}$ splits into a CLT term, an empirical-process term that cross-fitting makes $o_p(n^{-1/2})$, and the product-bias term of Theorem 15.4.

PART IV · MATHEMATICAL STATISTICS

15.12 Inference in Practice: Monte Carlo, Bootstrap and Overlap

Table 15.3. Monte Carlo behaviour of the estimators in the running example (Remark 15.2): 300 replications with $n = 1000$ (seeds 100–399), true ATE 3.5. Each sd is itself uncertain by about $\mathrm{sd}/\sqrt{598} \approx 4\%$.

Estimator	Mean	Bias	Sd
Naive difference	4.493	+0.993	0.0741
Outcome regression	3.503	+0.003	0.0486
IPW-HT, true e	3.501	+0.001	0.2670
IPW-HT, estimated $\hat{e}$	3.505	+0.005	0.0736
IPW-Hájek	3.503	+0.003	0.0566
AIPW	3.503	+0.003	0.0502

PYTHON LAB · `s15_examples.py` · `ex_sampling_distribution()`

Prints the table above; `coverage of the 95% influence-function CI of AIPW: 0.967`; `mean influence-function s.e.  of AIPW: 0.0535`. (With 300 replications a coverage has standard error $\sqrt{0.95 \cdot 0.05/300} = 0.013$, so 0.967 is compatible with 0.95.)

MATHEMATICAL STATISTICS · The nonparametric bootstrap for causal estimators

1 Resample n units with replacement. 2 *Refit every nuisance model* and recompute $\widehat{\mathrm{ATE}}^{*b}$. 3 Repeat B times; the percentile CI is $[\hat{q}_{0.025}, \hat{q}_{0.975}]$ of $\{\widehat{\mathrm{ATE}}^{*b}\}$ and $\widehat{\mathrm{se}}_{\mathrm{boot}}$ is their sd. Refitting matters: the bootstrap must see the variability of $\hat{e}$ and $\hat{\mu}_t$. For AIPW the bootstrap and the influence-function interval agree to first order (Theorem 15.5); for nearest-neighbour matching estimators the naive bootstrap is invalid (Abadie and Imbens, 2008).

Worked Example 15.10 (Bootstrap CI for AIPW)**.** On the $n = 4000$ running-example dataset (Remark 15.2), AIPW $= 3.524$. With $B = 300$ resamples (`bootstrap_aipw()`, seed 7), each refitting the logistic propensity and the linear outcome model, the bootstrap s.e. is the sd of the 300 replicates, 0.0245, and the percentile CI (2.5% and 97.5% quantiles of the replicates) is $[3.477, 3.570]$. The influence-function s.e. is $\mathrm{sd}(\hat{\varphi}_i)/\sqrt{4000} = 0.0258$, giving the Wald CI $3.524 \pm 1.96 \times 0.0258 = 3.524 \pm 0.051 = [3.473, 3.574]$. Both cover the sample ATE 3.506 and the population ATE 3.5.

PYTHON LAB · `s15_examples.py` · `ex_bootstrap_ci()`

```
boot s.e.  0.0245, CI [3.477, 3.570]; IF s.e.  0.0258, CI [3.473, 3.574]
```

Worked Example 15.11 (How weak overlap destroys precision)**.** Keep the running example (Remark 15.2) but let $e(x) = \mathrm{sigmoid}(\kappa(4x - 2))$; $\kappa = 1$ is the original design and larger κ means stronger confounding. For each κ we run 120 replications with $n = 1000$ (seeds 5000–5119); HT and Hájek use the estimated $\hat{e}$, and the bound is

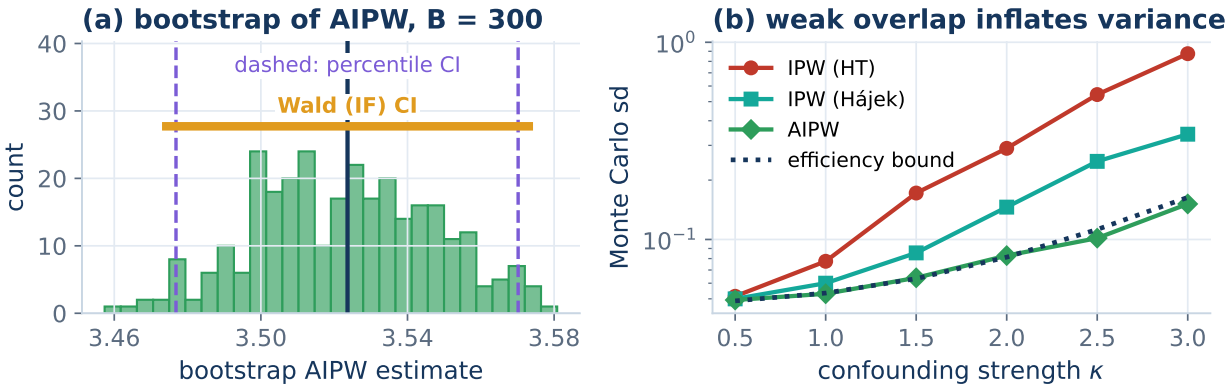


Figure 15.7. (a) Bootstrap distribution of AIPW ($B = 300$, Worked Example 15.10) with percentile (dashed) and Wald intervals. (b) Worked Example 15.11: as confounding strengthens, $e(x) = \text{sigmoid}(\kappa(4x-2))$ approaches 0 and 1 and the sd of HT explodes; AIPW follows the efficiency bound.

$\sqrt{V_{\text{eff}}(\kappa)/1000}$ from the formula of Worked Example 15.9 (integrated numerically). For $\kappa = 0.5, 1, 2, 3$ the smallest propensity $\text{sigmoid}(-2\kappa)$ is $0.269, 0.119, 0.018, 0.002$; the sd of HT is $0.052, 0.078, 0.290, 0.875$, of Hájek $0.050, 0.060, 0.146, 0.342$, of AIPW $0.049, 0.053, 0.083, 0.151$, against bounds $0.049, 0.053, 0.081, 0.163$. At $\kappa = 3$ AIPW seems to beat the bound ($0.151 < 0.163$), but with 120 replications an sd is uncertain by about $0.151/\sqrt{238} \approx 0.010$; a longer run with 1000 replications (`ex_bound_check()`) gives 0.1614 against the bound 0.1634. Trimming $\hat{e} \notin [\alpha, 1-\alpha]$ restores precision but changes the estimand.

PYTHON LAB · `s15_examples.py` · `ex_overlap_variance()`

```
kappa 3.0 (e in [0.002,0.998]):  sd HT 0.875, Hajek 0.342, AIPW 0.151, bound
0.163
ex_bound_check():  kappa 3.0, R = 1000:  sd AIPW 0.1614, bound 0.1634
```

PART IV · MATHEMATICAL STATISTICS

15.13 Worked Examples

Worked Example 15.12 (A confounded targeting simulation, all estimators checked against ground truth)**.** This is the running example of Remark 15.2: $n = 4000$ customers with engagement score $X \sim \text{Uniform}(0,1)$ (a confounder, playing the role of Z), true propensity $e(X) = \text{sigmoid}(4X-2)$ (highly-engaged customers are much more likely to receive a targeted offer T), baseline $Y(0) = 1+2X+\varepsilon_0$ and true heterogeneous treatment effect $\text{CATE}(x) = 2+3x$ (the offer helps more-engaged customers more), so the true $\text{ATE} = \mathbb{E}[2+3X] = 3.5$ and the sample ATE is 3.506. Figure 15.3 confirms adequate propensity-score overlap between treated and control groups ($\hat{e} \in [0.135, 0.878]$; a necessary condition for any of these estimators to be reliable — if some region of Z contained *only* treated units, no untreated comparison would exist there at all). Figure 15.4 compares the estimates: a naive difference in means (4.470, badly biased upward because engaged customers both received the offer more often *and* had higher baseline outcomes even without it — exactly Session 12's confounding pattern), outcome regression (15.4) (3.522), IPW (3.514 HT, 3.524 Hájek, both with the estimated $\hat{e}$), AIPW (15.5) (3.524, influence-function s.e. 0.026), and the coin-flip benchmark T^{rct} on the same

customers (3.533). All adjustment-based estimators land within statistical noise (about two s.e. of 0.026) of the sample ATE 3.506; only the naive comparison is off, by $4.470 - 3.506 = 0.964$, a relative bias of $0.964/3.506 = 27.5\%$ that would matter enormously for a real business decision.

PYTHON LAB · s15_examples.py · ex_targeting_simulation()

```
sample ATE 3.506; naive 4.470; reg 3.522; HT 3.514; Hajek 3.524; AIPW 3.524
(IF s.e.  0.026); RCT 3.533; naive bias 0.964 (27.5%); e-hat in [0.135,
0.878]
```

Worked Example 15.13 (Recovering who benefits most: the T-learner)**.** Using the same simulated data, fit two separate linear regressions of Y on engagement X — one using only treated customers, one using only untreated customers — and take their difference as $\widehat{\text{CATE}}(x) = 1.995 + 3.042x$ (intercepts and slopes of the two fits subtracted). At $x = 0$, 0.5, 1 this gives 1.995, 3.516, 5.036 against the truth 2, 3.5, 5; Fig. 15.5(a) shows that it tracks $\text{CATE}(x) = 2 + 3x$ closely across the full range of engagement scores. This is directly actionable: a firm with a limited marketing budget should target the *highest*-engagement customers first, not because they are easiest to reach, but because the estimated causal effect of the offer is largest for them — a decision that a single average treatment effect number could never have supported. (An additive S-learner $Y \approx a + bX + cT$ on the same data returns the constant $\hat{c} = 3.499$ for every x; see Worked Example 15.17.)

PYTHON LAB · s15_examples.py · ex_tlearner()

```
T-learner:  CATE-hat(x) = 1.995 + 3.042 x;    additive S-learner:  c = 3.499
for every x
```

Worked Example 15.14 (Why the RCT benchmark needs no identification argument at all)**.** In the simulated randomized experiment used as a benchmark above (T^{rct} of Remark 15.2), treatment was assigned by a fair coin flip *independent of* X: $e(x) = 0.5$ for everyone (mean engagement 0.509 vs 0.495 in the two arms, against 0.633 vs 0.364 in the observational data). Because assignment does not depend on X (or anything else), $Z = \emptyset$ trivially satisfies the backdoor criterion — there is no confounding to adjust for, so the naive difference in means *is* already the correct estimator of the ATE (3.533), with no need for the propensity or outcome models built above at all. This is the deepest connection this course draws between Sessions 12–15 and the classical A/B-testing framework familiar from any introductory statistics course: **a randomized experiment is not a different kind of causal inference — it is the special case where the identification problem Session 14 solved with real analytical work is instead solved automatically by the experimental design itself**, since randomization guarantees $e(Z)$ is known and constant by construction, making Z trivially sufficient for backdoor adjustment.

PYTHON LAB · s15_examples.py · ex_rct_benchmark()

```
RCT 3.533; mean X treated/control 0.509/0.495 (observational 0.633/0.364)
```

Worked Example 15.15 (What OLS with stratum dummies estimates — analytical)**.** Regress Y on T and a full set of stratum dummies. By Frisch–Waugh–Lovell the T-coefficient is $\hat{\beta} = \sum_i \tilde{T}_i Y_i / \sum_i \tilde{T}_i^2$ with $\tilde{T}_i = T_i - \hat{e}(z_i)$, $\hat{e}(z) = n_{1z}/n_z$. Within stratum z, writing $\hat{e}_z = \hat{e}(z)$,

the $n_{1z} = n_z\hat{e}_z$ treated units have $\tilde{T}_i = 1 - \hat{e}_z$ and the $n_{0z} = n_z(1-\hat{e}_z)$ controls have $\tilde{T}_i = -\hat{e}_z$, so

$$\sum \tilde{T}_i Y_i = n_{1z}(1-\hat{e}_z)\bar{Y}_{1z} - n_{0z}\hat{e}_z\bar{Y}_{0z} = n_z\hat{e}_z(1-\hat{e}_z)(\bar{Y}_{1z} - \bar{Y}_{0z}),$$
$$\sum \tilde{T}_i^2 = n_{1z}(1-\hat{e}_z)^2 + n_{0z}\hat{e}_z^2 = n_z\hat{e}_z(1-\hat{e}_z),$$

hence

$$\hat{\beta} = \frac{\sum_z w_z \hat{\tau}_z}{\sum_z w_z}, \qquad w_z = n_z\hat{e}_z(1-\hat{e}_z).$$

OLS weights strata by the *variance of treatment*, not by their size. In the hospital data of Worked Example 15.3, with $\hat{e}_z = 0.24, 0.45, 0.825$: $w = (100\cdot0.24\cdot0.76,\ 60\cdot0.45\cdot0.55,\ 40\cdot0.825\cdot0.175) = (18.24,\ 14.85,\ 5.775)$, and $\hat{\beta} = (18.24\cdot1 + 14.85\cdot2 + 5.775\cdot3)/(18.24+14.85+5.775) = 65.265/38.865 = 1.6793$ — neither the ATE (1.700) nor the ATT (2.107): the severe stratum, where almost everyone is treated, is down-weighted.

PYTHON LAB · `s15_examples.py` · `ex_regression_weights()`

```
weights = [18.24 14.85 5.775];  sum w tau = 65.265, sum w = 38.865;  formula = 1.6793; OLS coefficient on T = 1.6793
```

Additional Worked Examples (Instructor Supplement)

Worked Example 15.16 (Quantifying IPW's variance blow-up from a single extreme-propensity unit)**.** The overlap discussion of Section 15.4 is qualitative; here is a concrete instance. A treated unit i with $Y_i = 10$ has propensity $\hat{e}(z_i) = 0.05$ (rare: this unit's covariates made treatment unlikely). Its IPW contribution (before the factor $\frac{1}{n}$) is $T_iY_i/\hat{e}(z_i) = 10/0.05 = \mathbf{200}$. Compare to a "typical" treated unit with the same outcome $Y = 10$ but $\hat{e}(z) = 0.5$: its contribution is $10/0.5 = \mathbf{20}$. The rare-propensity unit alone contributes $10\times$ as much to the running sum $\frac{1}{n}\sum_i[\cdots]$ as a typical unit does — meaning a single observation can swing the entire average as much as ten ordinary ones combined. This is precisely the mechanism behind the variance inflation of Proposition 15.2 and Worked Example 15.11: near-violations of overlap do not merely add a little noise, they let individual data points dominate the estimator, which is why practitioners routinely trim or cap extreme propensity weights in applied work.

PYTHON LAB · `s15_examples.py` · `ex_extreme_propensity()`

```
Y/e with e = 0.05 :  200; Y/e with e = 0.50 :  20; ratio :  10
```

Worked Example 15.17 (Why an S-learner without interaction terms can miss heterogeneity entirely)**.** Two covariate strata, low and high engagement (coded $z = 0$ and $z = 1$), each half the population, have *true* effects CATE(low) $= \mu_1(\text{low}) - \mu_0(\text{low}) = 5 - 5 = 0$ and CATE(high) $= 5 - 2 = 3$ (so ATE $= 0.5(0) + 0.5(3) = 1.5$). Suppose an S-learner is fit with a purely *additive* functional form, $\hat{\mu}(z,t) = a + bz + ct$ — no $z \times t$ interaction term. Explain why this model cannot recover the true heterogeneous effects, and what it recovers instead.
Solution. With no interaction term, the model's estimated treatment effect $\hat{\mu}(z,1) - \hat{\mu}(z,0) = c$ is forced to be the *same constant for every* z — it structurally cannot output 0 for low-engagement customers and 3 for high-engagement customers simultaneously. Fit by least

squares against data generated from the true heterogeneous process, such a model converges to a weighted average of the two effects. Because z is binary, $a + bz$ is the same as a full set of stratum dummies, so by Worked Example 15.15 (in a large population, with $n_z/n \to P(z)$) $c = \sum_z w_z \mathrm{CATE}(z) / \sum_z w_z$ with weights $w_z = P(z)e(z)(1 - e(z))$. With equal propensities in both strata the weights are equal and $c = \mathrm{ATE} = 1.5$ exactly. With $e(\text{low}) = 0.2$, $e(\text{high}) = 0.5$: $w_{\text{low}} = 0.5 \cdot 0.2 \cdot 0.8 = 0.080$, $w_{\text{high}} = 0.5 \cdot 0.5 \cdot 0.5 = 0.125$, so $c = (0.080 \cdot 0 + 0.125 \cdot 3)/(0.080 + 0.125) = 0.375/0.205 = 1.829$ — correctly signed, yet erasing the fact that the true effect is 0 for one group and 3 for the other. This is the standard critique of S-learners: any regularization or model-family choice that shrinks or omits the treatment-covariate interaction will bias $\widehat{\mathrm{CATE}}(z)$ toward a pooled average, understating genuine heterogeneity — precisely the failure mode a T-learner (Worked Example 15.13) avoids by construction, since it fits two entirely separate models with no shared interaction term to shrink.

PYTHON LAB · `s15_examples.py` · `ex_slearner_additive()`

OLS fitted to an exact population of 10 000 units (5000 per stratum, outcomes equal to the stratum-arm means): `e(low) = 0.5, e(high) = 0.5: c = 1.500; e(low) = 0.2, e(high) = 0.5: weights 0.080, 0.125; c = 1.829`

Worked Example 15.18 (Why cross-fitting matters: an overfitting thought experiment)**.** Consider fitting an outcome model $\hat{\mu}_1$ on $n = 4$ treated units using a model flexible enough to have 4 free parameters (one per data point) — e.g., a lookup table. It achieves *zero* training residual by construction: $Y_i - \hat{\mu}_1(z_i) = 0$ for all 4 points. What happens to the AIPW correction term computed on this same data?

Solution. The correction term $\frac{T_i(Y_i - \hat{\mu}_1(z_i))}{\hat{e}(z_i)}$ is identically zero for every treated unit, since every in-sample residual is exactly zero by construction — *not* because the model has actually learned the true $\mu_1(z)$, but because it has simply memorized the training outcomes. (With outcomes $3, 5, 4, 7$ at $z = 0.1, 0.4, 0.6, 0.9$, a 1-nearest-neighbour lookup has in-sample residuals $0, 0, 0, 0$. Its leave-one-out residuals — each point predicted by its nearest neighbour among the other three — are $3 - 5 = -2$ (neighbour $z = 0.4$), $5 - 4 = 1$ ($z = 0.6$), $4 - 5 = -1$ ($z = 0.4$) and $7 - 4 = 3$ ($z = 0.6$).) AIPW would then behave as though the outcome model were perfectly correct (zero residuals mean the correction contributes nothing, exactly as when $\delta_1 \equiv 0$ in the appendix's proof), when in fact the fitted $\hat{\mu}_1$ may generalize terribly to any new unit. This is precisely why the double/debiased machine learning literature insists on **cross-fitting** (Algorithm 15.1): fit $\hat{\mu}_1, \hat{\mu}_0, \hat{e}$ on one part of the data and evaluate the correction terms on a *different* part, so that an overfit nuisance model cannot fool the estimator into reporting artificially small (and misleadingly reassuring) correction terms. Worked Example 15.6 shows the same effect numerically (0.458 in-sample vs 0.488 cross-fitted, truth 0.5).

PYTHON LAB · `s15_examples.py` · `ex_crossfit_overfit()`

`in-sample residuals: [0. 0. 0. 0.]; held-out (leave-one-out) residuals: [-2. 1. -1. 3.]`

PART V

Causal Effects in RL, Agents and Multi-Agent Systems

The same estimators under different names

PART V · AI CONNECTIONS

15.14 Off-Policy Evaluation in Reinforcement Learning

A contextual bandit logs (x_i, a_i, r_i) under a behaviour policy $\pi_0(a \mid x)$; we want the value $V(\pi) = \mathbb{E}_x \mathbb{E}_{a\sim\pi}[r]$ of a new policy π *without deploying it*. This is causal effect estimation with action = treatment, context = covariates, and π_0 = a *known* propensity:

$$\hat{V}_{\mathrm{IPS}} = \frac{1}{n}\sum_i \frac{\pi(a_i \mid x_i)}{\pi_0(a_i \mid x_i)} r_i, \qquad \hat{V}_{\mathrm{DR}} = \frac{1}{n}\sum_i \Big[\sum_a \pi(a \mid x_i)\hat{r}(x_i, a) + \frac{\pi(a_i \mid x_i)}{\pi_0(a_i \mid x_i)}\big(r_i - \hat{r}(x_i, a_i)\big)\Big].$$

The self-normalised version is $\hat{V}_{\mathrm{SNIPS}} = \sum_i w_i r_i / \sum_i w_i$ with $w_i = \pi(a_i \mid x_i)/\pi_0(a_i \mid x_i)$, and the direct method is $\hat{V}_{\mathrm{DM}} = \frac{1}{n}\sum_i \sum_a \pi(a \mid x_i)\hat{r}(x_i, a)$ for a reward model $\hat{r}(x, a) \approx \mathbb{E}[r \mid x, a]$. IPS is HT, SNIPS is Hájek, DM is outcome regression and DR is AIPW (Dudík, Langford and Li, 2011). Positivity becomes *exploration*; in sequential RL the weights become products $\prod_t \pi(a_t \mid s_t)/\pi_0(a_t \mid s_t)$ whose variance grows with the horizon (Sutton and Barto, 2018, Ch. 5).

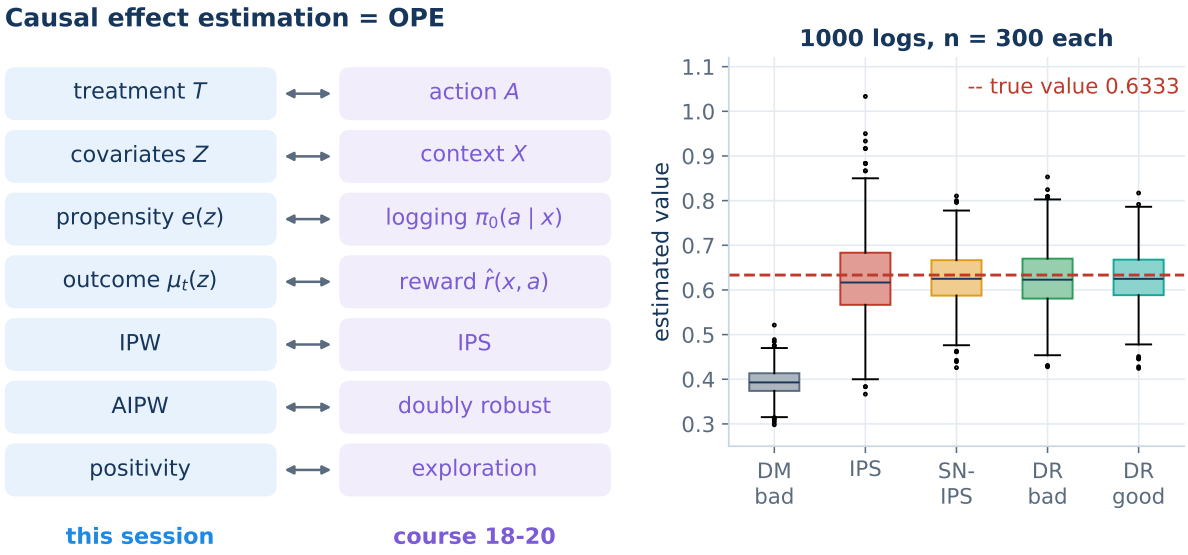


Figure 15.8. Left: the dictionary between causal effect estimation and off-policy evaluation. Right: 1000 simulated logs ($n = 300$ each) of the three-context, three-action bandit of Worked Example 15.19; "bad" and "good" refer to the context-free and the saturated reward model. A misspecified reward model makes DM badly biased; DR repairs it; with a good reward model DR equals DM.

Worked Example 15.19 (Doubly robust off-policy evaluation)**.** Contexts x_1, x_2, x_3 are equally likely; actions are a_1, a_2, a_3; rewards are Bernoulli with success probabilities $r(x, a)$, and the logging policy $\pi_0(a \mid x)$ is known (`OPE` in the script):

	$r(x,a)$			$\pi_0(a \mid x)$			target
	a_1	a_2	a_3	a_1	a_2	a_3	$\pi(x)$
x_1	0.3	0.5	0.4	0.6	0.2	0.2	a_2
x_2	0.6	0.2	0.5	0.2	0.6	0.2	a_1
x_3	0.2	0.4	0.8	0.5	0.3	0.2	a_3

The logging policy is worth $\sum_a \pi_0(a \mid x) r(x,a) = 0.36, 0.34, 0.38$ in the three contexts (e.g. $0.6 \cdot 0.3 + 0.2 \cdot 0.5 + 0.2 \cdot 0.4 = 0.36$), on average 0.36; the deterministic target policy is worth $(0.5 + 0.6 + 0.8)/3 = 0.6333$. We simulate 1000 logs of $n = 300$ rounds each (seed 99). With a misspecified reward model that ignores the context ($\hat{r}(x,a) =$ mean logged reward of action a): DM has bias -0.2396 (RMSE 0.2415); IPS bias -0.0048, sd 0.0972; SNIPS bias -0.0061, sd 0.0598; DR bias -0.0076, sd 0.0655. (These small biases are shared Monte Carlo noise: all estimators use the same 1000 logs, and $0.0972/\sqrt{1000} = 0.003$.) With a saturated reward model ($\hat{r}(x,a) =$ the mean reward in cell (x,a)) DR coincides with DM in every log (sd 0.0592): for each context the residuals $r_i - \hat{r}(x_i, a_i)$ of the rounds that played the target action sum to zero, so the DR correction vanishes — the bandit analogue of Worked Example 15.4, where a saturated nuisance fit made two estimators coincide.

PYTHON LAB · s15_examples.py · ex_ope_bandit()

```
logging 0.3600; true 0.6333; DM bias -0.2396, RMSE 0.2415; IPS bias -0.0048,
sd 0.0972; SNIPS -0.0061, 0.0598; DR -0.0076, 0.0655; DR-good = DM-good sd
0.0592
```

LINK TO COURSE SESSION 18 · Dynamic Programming and Q-Learning

Off-policy learning and evaluation share one problem: data come from a behaviour policy. Ratios π/π_0 are IPW weights, and ϵ-greedy exploration guarantees positivity, $\pi_0(a \mid s) \geq \epsilon/|\mathcal{A}|$.

LINK TO COURSE SESSION 20 · Deep and Model-based RL

Offline RL learns only from logs, so every improvement step is a causal extrapolation; doubly robust OPE (learned reward model + importance weights) validates a policy before deployment, and distribution shift is a positivity violation.

PART V · AI CONNECTIONS

15.15 Agentic AI: A/B Tests and Uplift Routing

Comparing two versions of an LLM agent on randomized traffic is an RCT, analysed with the Neyman CI of Proposition 15.3. With *logged* traces, where users or heuristics chose the version, task type and user segment confound the comparison and AIPW or DML with $Z =$ trace features is needed. The CATE answers: *for which queries is the expensive version worth it?*

Worked Example 15.20 (Routing between agent versions)**.** A hypothetical A/B test of agent v1 (single LLM call) against v2 (tool use + reflection, cost $c = 0.05$ success units per query) gives success rates v1/v2 of 0.80/0.82 (factual queries, 40% of traffic), 0.55/0.70 (reasoning, 35%) and 0.50/0.75 (coding, 25%). The CATEs (v2 minus v1) are $0.82 - 0.80 = 0.02$,

$0.70 - 0.55 = 0.15$, $0.75 - 0.50 = 0.25$, and the ATE is $0.40 \cdot 0.02 + 0.35 \cdot 0.15 + 0.25 \cdot 0.25 = 0.008 + 0.0525 + 0.0625 = 0.1230$. Net value per query:

- always v1: $0.40 \cdot 0.80 + 0.35 \cdot 0.55 + 0.25 \cdot 0.50 = 0.32 + 0.1925 + 0.125 = 0.6375$;
- always v2: $0.40 \cdot 0.82 + 0.35 \cdot 0.70 + 0.25 \cdot 0.75 - 0.05 = 0.7605 - 0.05 = 0.7105$;
- route to v2 only where CATE $> c = 0.05$ (reasoning and coding): $0.40 \cdot 0.80 + 0.35 \cdot (0.70 - 0.05) + 0.25 \cdot (0.75 - 0.05) = 0.32 + 0.2275 + 0.175 = 0.7225$,

better than either fixed choice (Fig. 15.9).

PYTHON LAB · `s15_examples.py` · `ex_agent_routing()`

```
CATE 0.02, 0.15, 0.25; ATE 0.1230; v1 0.6375, v2 0.7105, routing 0.7225
```

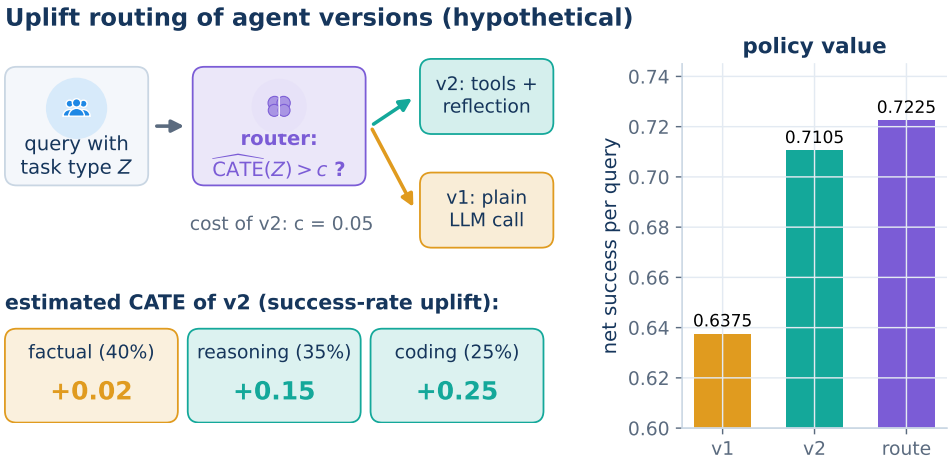


Figure 15.9. Uplift-based routing in an agentic system: a router estimates the CATE of the expensive agent version from the task type and sends a query there only when the uplift exceeds its cost.

LINK TO COURSE SESSION 26 · Agentic AI

Tool use, memory and reflection are *treatments*. An agent deciding whether to call a tool executes a policy $\pi(z)$: evaluating it on logs is OPE, improving it is policy learning with AIPW scores, and its router is an uplift model.

PART V · AI CONNECTIONS

15.16 Multi-Agent AI: When SUTVA Fails

SUTVA's "no interference" clause is violated whenever units interact: a treated agent changes its neighbours' outcomes through communication, competition or shared resources. Write $Y_i = f(T_i, \bar{T}_{N(i)})$ with $\bar{T}_{N(i)}$ the share of treated neighbours (an *exposure mapping*). The *direct* effect $f(1, s) - f(0, s)$ differs from the *total* effect $f(1, 1) - f(0, 0)$ of treating everyone, which is what a deployment decision needs.

Worked Example 15.21 (Spill-overs among agents on a ring)**.** 400 agents sit on a ring, each interacting with two neighbours on each side, so $\bar{T}_{N(i)}$ is the share of treated among four neighbours; $Y_i = 1 + 1.0\,T_i + 1.5\,\bar{T}_{N(i)} + \varepsilon_i$ with $\varepsilon_i \sim N(0, 1)$. The direct effect is 1.0 and the

total effect is $f(1,1) - f(0,0) = 1.0 + 1.5 = 2.5$. Under independent Bernoulli(1/2) assignment, $\bar{T}_{N(i)}$ is independent of T_i with mean 1/2 in both arms, so the difference in means estimates only the direct effect, $1.0 + 1.5(\frac{1}{2} - \frac{1}{2}) = 1.0$: mean 0.991 over 500 replications (seed 5). Randomizing *blocks* of $m = 20$ adjacent agents (each block treated with probability 1/2) makes neighbours share treatment. In a treated block, the agent at each end has two of its four neighbours outside the block, each treated with probability 1/2, so $\mathbb{E}\bar{T}_{N(i)} = (2 + 2 \cdot \frac{1}{2})/4 = \frac{3}{4}$; the second agent from each end has one outside neighbour, $\mathbb{E}\bar{T}_{N(i)} = (3 + \frac{1}{2})/4 = \frac{7}{8}$; the other $m - 4$ agents have $\bar{T}_{N(i)} = 1$. The block average is $[(m-4) + 2 \cdot \frac{7}{8} + 2 \cdot \frac{3}{4}]/m = 1 - \frac{0.75}{m}$, and by symmetry it is $\frac{0.75}{m}$ in a control block. Hence $\mathbb{E}[\text{DiM}] \approx 1 + 1.5\big(1 - \frac{1.5}{m}\big) = 1 + 1.5\big(1 - \frac{3}{2 \cdot 20}\big) = 1 + 1.5 \cdot 0.925 = 2.3875$, and the simulation gives mean 2.390. The remaining gap to 2.5 comes from the agents at block boundaries.

PYTHON LAB · `s15_examples.py` · `ex_interference()`

```
Bernoulli design:  mean 0.991;    cluster design (blocks of 20):  mean 2.390;
theory 2.3875
```

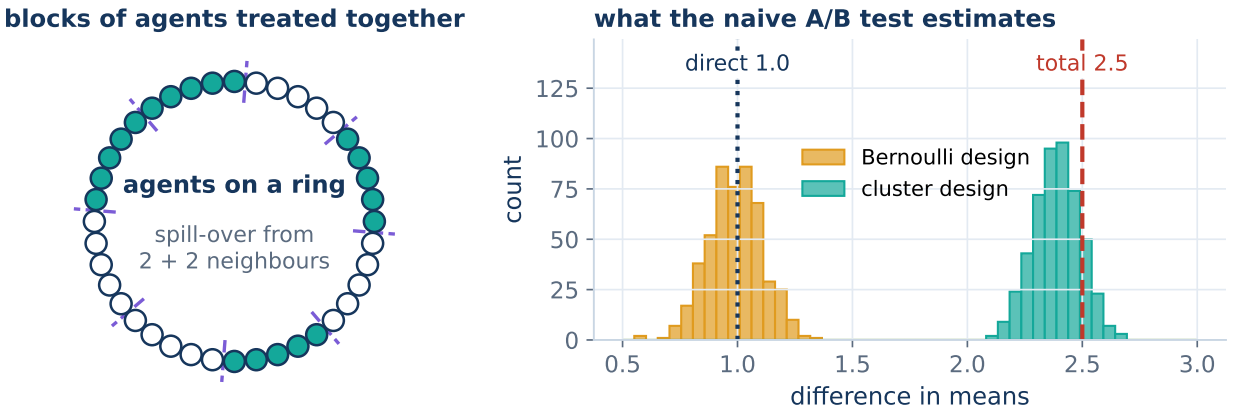


Figure 15.10. Interference among agents. Left: block (cluster) randomization on a ring (schematic: 40 agents in blocks of 5; the simulation of Worked Example 15.21 uses 400 agents in blocks of 20). Right: 500 simulated experiments; the naive A/B test recovers only the direct effect under independent assignment, and nearly the total effect under cluster assignment.

LINK TO COURSE SESSION 27 · Multi-Agent AI

Evaluating a new policy for *one* agent while the others keep theirs measures a direct effect; deploying it to *all* agents changes the equilibrium (a total effect). Cluster randomization and exposure mappings are the causal tools for coordination experiments.

AI CONNECTION · Trustworthy AI

Counterfactual fairness (Kusner et al., 2017) asks that a prediction for individual i would be the same had i's protected attribute been different — a rung-3 counterfactual that, by Proposition 15.1, needs structural assumptions beyond the data. CATE estimates also expose *who* is harmed by a policy, a basic audit requirement; fairness audits and "the decision would change if…" explanations (course session 31) are causal effect estimates resting on today's four assumptions.

AI CONNECTION · Deep Learning and Generative Models

Two-headed networks such as TARNet (Shalit, Johansson and Sontag, 2017) learn a shared representation $\Phi(z)$ and outcome heads h_0, h_1 — a neural T-learner with an S-learner's data sharing; adding a propensity head and an AIPW-style loss gives doubly robust training. Representation balancing penalises distance between $\Phi(Z) \mid T{=}1$ and $\Phi(Z) \mid T{=}0$, an echo of the balancing property (course sessions 3–4).

Table 15.4. This session in the Advanced AI course.

Concept today	Reappears as	Where
Propensity $e(z)$	behaviour / logging policy $\pi_0(a \mid x)$	course 18–20
IPW / Hájek	importance-sampling OPE, IPS / SNIPS	course 18–20
AIPW, orthogonal scores	doubly robust OPE, DR policy learning	course 20, 26
CATE and policy rule	uplift routing, tool-use decisions	course 26
SUTVA / interference	spill-overs among interacting agents	course 27
ITE bounds, counterfactuals	counterfactual fairness and explanations	course 31
Balancing, CATE heads	TARNet-style deep causal models	course 3–4

PART V · AI CONNECTIONS

15.17 Summary

Table 15.5. Cheat-sheet of key formulas.

Object	Formula
Identification	$\mathbb{E}[Y(t)] = \mathbb{E}[\mu_t(Z)] = \mathbb{E}[\mathbb{1}\{T{=}t\}Y/P(T{=}t \mid Z)]$
ATT	$\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[\mu_0(Z) \mid T{=}1]$
AIPW score	$\varphi = \mu_1 - \mu_0 + \frac{T(Y-\mu_1)}{e} - \frac{(1-T)(Y-\mu_0)}{1-e}$
AIPW bias	$\mathbb{E}[(\hat{\mu}_1 - \mu_1)(\hat{e} - e)/\hat{e}]$ (product of errors)
Efficiency bound	$\mathbb{E}[\sigma_1^2/e + \sigma_0^2/(1-e) + (\tau - \text{ATE})^2]$
DML (PLM)	$\hat{\theta} = \sum \tilde{T}\tilde{Y} / \sum \tilde{T}^2$, cross-fitted residuals
Policy value	$V(\pi) = \mathbb{E}[(\tau(Z) - c)\pi(Z)]$, optimum $\mathbb{1}\{\text{CATE} > c\}$
Benefit bounds	$\max(0, p_1 - p_0) \le P(\text{benefit}) \le \min(p_1, 1 - p_0)$

Table 15.6. Common pitfalls and their fixes.

Pitfall	Fix
Reading the OLS coefficient of T as the ATE	regression/AIPW with interactions; know the $e(1-e)$ weights
Ignoring poor overlap	plot $\hat{e}$ by arm; trim and state the new estimand
Evaluating nuisance fits in-sample	cross-fitting (DML) or honesty (forests)
Interpreting CATE as an individual effect	CATE averages over a profile; ITEs are only bounded
A/B testing interacting agents unit by unit	cluster randomization, exposure mappings
Adjusting for post-treatment variables	Z must be pre-treatment (no descendants of T)

THINK ABOUT IT **An agent's logs record which tool it called and whether the user was satisfied. Which variables must you log so that ignorability can hold for "what if it always called tool A"?**

THINK ABOUT IT **Why can IPW with an estimated propensity be more precise than IPW with the true one?**

THINK ABOUT IT **If two agents learn from each other, what is the "unit" of a randomized experiment?**

Further reading. Hernán and Robins (2020), Parts I–II; Imbens and Rubin (2015); Pearl (2009), Ch. 3, 11; Wasserman (2004), Ch. 8, 16; Rosenbaum and Rubin (1983); Cochran (1968); Abadie and Imbens (2006, 2008); Hirano, Imbens and Ridder (2003); Robins, Rotnitzky and Zhao (1994); Hahn (1998); Chernozhukov et al. (2018); Künzel et al. (2019); Nie and Wager (2021); Wager and Athey (2018); Dudík, Langford and Li (2011); Sutton and Barto (2018).

APPENDIX

15.18 Appendix: Proofs

Proof sketch of double robustness (Theorem 15.3)

Write $\hat{\mu}_1(z) = \mu_1(z) + \delta_1(z)$ where μ_1 is the true outcome function and δ_1 is the model's error; similarly $\hat{e}(z) = e(z) + \eta(z)$. The AIPW estimand's expectation (for the treated-arm contribution, by symmetry) is

$$\mathbb{E}\left[\hat{\mu}_1(Z) + \frac{T(Y - \hat{\mu}_1(Z))}{\hat{e}(Z)}\right] = \mathbb{E}\left[\mu_1(Z) + \delta_1(Z)\Big(1 - \frac{e(Z)}{\hat{e}(Z)}\Big)\right],$$

using $\mathbb{E}[T \mid Z] = e(Z)$ and $\mathbb{E}[Y \mid T{=}1, Z] = \mu_1(Z)$ (backdoor unconfoundedness). If $\delta_1(Z) \equiv 0$ (outcome model correct), the bias term vanishes regardless of η. If instead $\eta(Z) \equiv 0$ (propensity model correct, so $\hat{e} = e$), the factor $\big(1 - \frac{e(Z)}{\hat{e}(Z)}\big) = \big(1 - \frac{e(Z)}{e(Z)}\big) = 0$ regardless of δ_1. Either condition alone kills the bias term, leaving $\mathbb{E}[\mu_1(Z)] = \mathbb{E}[Y \mid \mathrm{do}(T{=}1)]$

by the backdoor formula — proving consistency under either (but not necessarily both) correct specifications. The full proof, with the analogous control-arm term and a formal treatment of the estimator's asymptotic variance (which is minimized when *both* models are correctly specified, an efficiency bonus beyond mere robustness), is standard in the semiparametric efficiency literature (Robins, Rotnitzky & Zhao, 1994; Chernozhukov et al., 2018, on "double/debiased machine learning," which extends exactly this idea to allow $\hat{\mu}$ and $\hat{e}$ to be arbitrary flexible ML models via sample-splitting). Theorem 15.4 is the exact version of this argument: $\delta_1(1 - e/\hat{e}) = \delta_1\eta/\hat{e}$.

Proof sketch of the CLT for cross-fitted AIPW (Theorem 15.5)

Let P_n denote the empirical average over a fold of size n_k, $\eta = (\mu_0, \mu_1, e)$ the true nuisances and $\hat{\eta}$ those fitted on the other folds; write $m(W;\eta)$ for the uncentred AIPW score so that $\psi = \mathbb{E}\, m(W;\eta)$. Then

$$P_n m(W;\hat{\eta}) - \psi = \underbrace{(P_n - P)\, m(W;\eta)}_{\text{(i)}} + \underbrace{(P_n - P)\big[m(W;\hat{\eta}) - m(W;\eta)\big]}_{\text{(ii)}} + \underbrace{P\big[m(W;\hat{\eta})\big] - \psi}_{\text{(iii)}}.$$

(i): by the classical CLT, $\sqrt{n}\,\text{(i)} \xrightarrow{d} N(0, V_{\text{eff}})$. **(ii)**: conditionally on the training folds, $\hat{\eta}$ is fixed, so (ii) is a mean-zero average with conditional variance $\|m(\cdot;\hat{\eta}) - m(\cdot;\eta)\|_2^2/n_k$; strict overlap and L_2-consistency make this $o_p(1/n)$, hence (ii) $= o_p(n^{-1/2})$ by Chebyshev. Without cross-fitting this step would require a Donsker condition that flexible ML fits violate. **(iii)** is exactly the product bias of Theorem 15.4, bounded by $\varepsilon^{-1}\sum_t \|\hat{\mu}_t - \mu_t\|_2 \|\hat{e} - e\|_2 = o_p(n^{-1/2})$. Averaging over the K folds and multiplying by $\sqrt{n}$ yields the result; consistency of $\hat{V}$ follows from the same decomposition applied to $\hat{\varphi}_i^2$.

CONNECTIONS AND TAKEAWAYS

The notes' arc is now complete. Sessions 5–11 built a language for representing and computing with joint distributions — conditional independence, graphs, exact and approximate inference, and learning. Session 12 showed, with real numbers, that this language cannot by itself answer questions about intervention. Session 13 gave arrows a stronger, mechanistic meaning (SCMs) that supports exactly those questions. Session 14 showed when and how a causal effect can be recovered from observational data and a graph alone (identification). And today closes the loop back to practice: once identification tells us *what* to estimate, the machine learning tools students already know — regression, classification, cross-fitting — tell us *how well* we can estimate it, and the special case of a randomized experiment shows that everything built since Session 12 was, all along, a way of solving — analytically, from observational data — the very problem that random assignment solves for free by design.

Course-level takeaway: causal inference is not a separate discipline bolted onto machine learning. It is machine learning aimed, by a graph-theoretic argument, at a precisely defined and provably estimable target — and knowing when that target is estimable at all, which this course's entire second half was devoted to establishing, is exactly the judgment a working data scientist needs before ever fitting a single model.

Next: course session 17 opens reinforcement learning, where action = treatment, behaviour policy = propensity and exploration = positivity.